Endogenous Strength in Competition Between and Within Groups*

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Abstract

Groups often compete for indivisible rewards or for divisible rewards without first establishing internal allocation rules. In such cases, members of a group must compete with one another. An individual's strength in competition with other group members may be exogenously determined, or it may depend on how much she contributes to the group's efforts. In this paper, I explore the link between intergroup competition and strength in intragroup competition. I develop a model of a two-stage contest in which groups compete for an indivisible prize in the first stage and members of the winning group compete with one another in the second stage to determine the allocation of the prize. I explore different functions relating group contribution and individual strength. I then design and conduct an experiment in which subjects play the two-stage contest with different strength functions. I find that total contributions remain constant across treatments, but the timing of contributions differs. I also find that subjects compete differently in response to asymmetry depending on how asymmetry arises. Individuals care about whether advantages are given or earned and especially about *how* advantages are earned. The results shed light on optimal organizational or industrial policy in situations in which groups and individuals compete.

Keywords: group behavior, contests, experiment, personnel economics

JEL: C72, C91, C92, M54

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1 Introduction

In many situations, groups compete for an indivisible prize or for a divisible prize without previously establishing rules for sharing internally. In such cases, group members work collectively against other groups and then compete internally over prize allocation. When it comes time to compete internally, some group members may possess an advantage over others. In organizations, political alliances, R&D groups, and sports teams an individual's relative strength in internal conflict sometimes depends on exogenous factors like natural ability or seniority. Other times, relative strength depends on endogenous factors. In many cases, a group member's position in competition with other group members depends on how much she contributes to the group's collective effort. With this environment in mind, I develop a model and conduct a laboratory experiment to study competition between and within groups. I specifically explore the link between group contribution and individual strength and study how varying the nature of this link affects behavior.

As an example of the environment, consider a professional services firm in which workers coordinate to attain a client and then compete internally over who manages the new client's account. An employee's strength in intra-firm conflict may depend on how much effort she contributes to attaining the client initially. In some cases, pursuing a client increases an employee's understanding of the client's needs and gives her an advantage in competition against peers. In other cases, exerting effort for the sake of the firm may induce burnout and limit a worker's chance of winning client management. This and many other real-world examples illustrate an environment in which competition takes place between and within groups where some link may exist between group contribution and individual strength.

To study this setting, I model a two-stage contest in which groups compete in the first stage, and individual members of the winning group compete with each other in the second stage. Agents maximize expected payoffs by choosing contributions in each stage from an endowment. In the individual stage, contributions are multiplied by a strength parameter. The main innovation of the model is the link between group stage contributions and strength. I consider different functions relating group stage contribution and strength and characterize subgame-perfect equilibria. I then compare equilibrium contributions when the strength is increasing versus decreasing in group stage contributions. As benchmarks, I also consider when strength is fixed and symmetric or exogenously assigned and random. I explore comparative statics on contributions in each stage as well as measures of aggregate expenditures across both stages. The theoretical results provide testable hypotheses for an experiment.

In my experiment, two groups of two subjects each play the two-stage contest. Subjects receive an endowment of points from which they select group stage contributions and individual stage contributions. I randomly rematch groups each round for 20 repeated rounds. Subjects participate in one of four treatments in a between-subjects design each consisting of a different relationship between group stage contribution and individual strength. I run two treatments in which strength is endogenously determined by group stage contributions and two treatments in which strength is exogenously assigned. In the **Increasing** treatment, higher group stage contributions increase strength, while in the **Decreasing** treatment, higher group stage contributions decrease strength. In the **Symmetric** treatment, each subject possesses a fixed and symmetric strength, while in the **Random** treatment, strengths are randomly drawn from a uniform distribution. I collect data on contributions in each stage and compare across treatments.

I observe three main sets of results. First, I find that, contrary to the bulk of previous

research on laboratory experimental contests, subjects in the individual stage do not typically overinvest on average relative to equilibrium predictions. Most contest experiments involve subjects playing one-shot or repeated contests in isolation. Perhaps embedding contests in more complex dynamic settings deters overinvestment.

Next, I find that subjects who face an increasing strength function contribute significantly more in the group stage and significantly less in the individual stage compared to subjects who face a decreasing strength function. The differences within each stage cancel out so that aggregate expenditures across both stages do not differ significantly across treatments. In other words, total contributions over the group contest and the individual contest are the same across treatments, but the timing and type of contributions differ. These results contradict theoretical predictions. If any and all rent-seeking activity is wasteful from society's perspective, then the cases I study in my experiment should be considered equally inefficient. However, situations may exist in which collective and/or individual effort is actually considered productive. The productivity or wastefulness of different types of effort may be an artifact of the environment or a prerogative of a policymaker or manager.

In some cases, contributions to collective effort may be considered productive while effort expended in internal conflict may be considered wasteful. For example, a firm profits when employees coordinate to attain clients or resources but suffers from infighting over client assignment or resource allocation. According to my results, managers should pursue a policy which rewards group contributions with higher individual strength. Somewhat counterintuitively, by incentivizing higher group contributions through the promise of higher strength in internal competition, the firm actually discourages in-fighting without sacrificing collective effort.

There may be other cases in which a principal values individual effort but considers group effort wasteful. For example, research and development and patent races involve much duplication of effort between teams or groups, but the valuable innovation occurs through internal competition. In this case, it is optimal to minimize duplication of efforts across groups and promote competition within any one particular group. According to my experimental data, an optimal policy in this case would involve a decreasing strength function which disincentivizes group effort but promotes competition between individual group members. Overall, my first main set of results illustrates how, contrary to theoretical predictions, optimal managerial or industrial policy may depend on whether group effort or individual effort is considered productive versus wasteful.

The final main set of results focuses on the individual stage contest. In the individual stage in my experiment, subjects participate in a standard lottery contest with asymmetry. The key difference from past studies is that the asymmetry may arise endogenously based on choices from the group stage. When the strength function is increasing, an advantaged player in the individual stage earns her position by out-contributing her group mate. On the other hand, when the strength function is decreasing, an advantaged player in the individual stage earns her position by shirking. In the two other treatments, strengths are exogenously assigned, either symmetrically or asymmetrically within groups. Theoretically, given some vector of strengths, subjects in a contest should behave identically regardless of how strengths are assigned. An advantaged subject shouldn't care whether she earned that position by hard work, shirking, or through random assignment. Variation in contributions should be fully explained by variation in relative strengths between contestants. Any remaining variation must be due to some behavioral biases towards particular ways in which strengths are assigned.

In my experiment, I indeed find that individuals respond differently to asymmetry depend-

ing on how that asymmetry arises. Individual stage contributions vary across treatments even when controlling for measures of asymmetry. Diving deeper into treatment differences, I observe that individuals compete significantly more fiercely in an asymmetric contest when higher strength comes from shirking. Furthermore, as strengths get more uneven, competition gets even fiercer under decreasing endogenous strength but milder under increasing endogenous strength or random assignment. I see this pattern reflected in my data both in terms of raw individual stage contributions and the stronger player's share of contributions in the contest. Together my results illustrate a pattern that individuals appear much more willing to accept the advantage afforded to stronger players when strength is earned through high group contribution or randomly assigned compared to when it is earned through low group contribution. This behavioral phenomenon resembles the entitlement effect observed in the literature on other-regarding preferences whereby people behave less generously when they earn the right to propose offers in dictator and ultimatum games. In my competitive environment, individuals care not only about whether an advantaged position is given or earned but also about *how* the advantaged position is earned. This nuanced result expands the previous findings on entitlement effects and earned versus assigned property rights.

My data suggest that competitiveness varies in asymmetric situations depending on how asymmetry comes about. With this in mind, managers and policymakers over unequally matched agents must pay special attention to how agents earn positions of power and whether advantages come from exogenous factors, outworking others, or shirking. I now highlight some of the related literature I build on in this paper.

2 Related Literature

This paper contributes to previous research on group and individual contests, contest design and personnel economics, and behavioral biases. Substantial theoretical and experimental research studies group contests.[1] Münster (2009) extends the axiomatic characterization of contest success functions by Skaperdas (1996) and Clark & Riis (1998) to contests between groups. Experiments that study rent-seeking in groups explore a variety of topics including different production functions aggregating group efforts (Sheremeta, 2011; Cason et al., 2012), asymmetry between or within groups (Bhattacharya, 2016; Fallucchi et al., 2021; Brookins et al., 2015) or the effects of communication (Cason et al., 2012, 2017). Other studies consider the effects of group sizes and compare group contests to contests between individuals or between individuals and groups (Abbink et al., 2010; Ahn et al., 2011). In my experiment an individual contest is embedded within a group contest dynamically, and group contest contributions sometimes affect the nature of the individual contest. I study how this link between group and individual incentives affects competition in each contest.

I also contribute to work on individual contests with asymmetry. Early models of individual rent-seeking games (Tullock, 1967, 1980; Rogerson, 1982; Skaperdas, 1996) sparked a vast theoretical and experimental literature.[2] One branch of this research focuses on contests with asymmetric players (Stein, 2002; Baik, 1994; Schotter & Weigelt, 1992; Kirkegaard, 2012; Franke et al., 2013). Different models incorporate asymmetry through costs, valuations, or productivities. In my model, I follow one of the approaches modeled by Baik (1994) and Stein (2002) and

[1] See Sheremeta (2018) for a review of group contests.
[2] See Dechenaux et al. (2015) for a review of individual contests.

assign each agent a unique strength parameter which enters multiplicatively in the contest success function. Fonseca (2009) studies this type of asymmetry experimentally and explores how behavior varies in symmetric versus asymmetric contests. The paper finds that expenditures decrease in asymmetric contests for both the stronger and the weaker players and that both invest at about the same level on average. The basic framework of the individual stage in my experiment resembles Fonseca (2009), so I reference it frequently when analyzing my results.

I also contribute to previous research on dynamic contests. In particular, I build upon previous theoretical and experimental work studying multi-stage contests. Previous models and experiments primarily focus on multi-stage contests with carryover (Schmitt et al., 2004; Sheremeta, 2010a; Ryvkin, 2011) or elimination tournaments (Gradstein & Konrad, 1999; Altmann et al., 2012; Sheremeta, 2010a,b). This paper most readily contributes to a small strand of theoretical research that studies two-stage contests in which groups and individual group members compete (Wärneryd, 1998; Katz & Tokatlidu, 1996). Analysis of these models focuses on the effects of varying group sizes and numbers of groups or the timing of competition (Münster, 2007). Extensions of the basic model study situations when asymmetry exists within or between groups (Choi et al., 2016; Stein & Rapoport, 2004).

While much theoretical work has studied this two stage environment, little empirical research exists. Most similarly to my paper, Bhattacharya & Rampal (2024) study an intragroup contest followed by an intergroup contest. Groups first establish an internal prize allocation rule before competing with other groups. I build on this environment by reversing the order of the stages and establishing an explicit link between them. In my experiment, groups compete first and then individual group members compete internally to determine prize allocation. I am the first to impose a link between the group and individual stages and study the effects on expenditures. In doing so, I develop a richer and more realistic environment to capture the tension between group and individual incentives.

The insights of my model and results expand our understanding of rent-seeking in various political economic, industrial and organizational settings. In particular, I contribute to research on personnel economics and contest design which informs organizations and managers. Lazear & Rosen (1981) provide the seminal work on contests in a labor setting and inspire many extensions. Most relevant to my paper, Tsoulouhas et al. (2007) investigate optimal promotion contests for heterogeneous candidates (namely, insiders versus outsiders), and Fu & Lu (2012) discuss the optimal design of multi-stage contests. Building on this literature, in the analysis of my results, I consider optimal policies for firms in different situations involving both external and internal competition.

Finally, I contribute to previous work studying the behavioral impacts of the assignment of property rights. Hoffman et al. (1994) studies how other-regarding preferences differ in dictator and ultimatum games when the right to be the first-mover is given versus earned. This paper discovers an entitlement effect whereby people who earn their position as proposer behave less generously compared to people who receive their position by luck or random assignment. Follow up research largely confirms the entitlement effect (Cherry et al., 2002; Cappelen et al., 2007; Oxoby & Spraggon, 2008). Many of these studies follow a similar multi-stage structure as my experiment where a preliminary stage may be used to determine one's position in the main stage. The main insight in this literature is that individuals care about how advantages and property rights are assigned and modify their behavior accordingly even when economic theory suggests that such contextual factors shouldn't matter. I expand the research of entitlement effects to a new competitive setting and explore how it manifests in contests where advantages

are earned or given. I also consider that individuals may behave differently depending on exactly *how* advantages are earned. I study how this distinction impacts competitiveness and add new detail to our understanding of behavioral entitlement effects.

In the following section, I present the model and analysis highlighting different links between group contributions and individual strength. I then describe the experimental design and present three sets of key findings.

3 Model

Consider two groups $g \in \{a, b\}$ of two risk-neutral agents $i \in \{1, 2\}$ who compete in a two-stage contest for a known and common-value prize $V > 0$. Each agent ig receives an endowment $w > 0$ from which she may make contributions over both stages of the two-stage contest. In the first stage, the group stage, the two groups compete in a group contest. One group wins and advances to the second stage, the individual stage, in which members of the winning group compete for the prize. In the individual stage, each agent possesses strength θ_{ig} which may be exogenously assigned or endogenously dependent on group stage contributions.

3.1 Group Stage

In the group stage, two groups compete for the right to advance to the individual stage and compete for a prize. Each agent chooses how much of her endowment to contribute to the group contest. Individuals simultaneously select group stage contribution x_{ig}, and a group's contribution is the sum of its members' group stage contributions $X_g = x_{1g} + x_{2g}$. Any portion of an agent's endowment that she doesn't contribute in the group stage, $w - x_{ig}$, remains with her if she advances to the individual stage. Once both groups submit their contributions, one winning group is determined stochastically. According to a group Tullock lottery contest success function (Münster, 2009; Tullock, 1980), group g wins with the following probability:

$$p_g = \begin{cases} \frac{X_g}{X_a + X_b} & \text{if} \quad X_a + X_b > 0 \\ \frac{1}{2} & \text{otherwise} \end{cases} \tag{1}$$

Agents earn no reward in the group stage. Members of the winning group advance to the individual stage where they have the opportunity to compete against one another for the prize V.

3.2 Individual Stage

In the individual stage, each member of the winning group chooses how much of her remaining endowment to contribute to the individual contest. Individuals simultaneously select individual stage contributions y_{ig}. Individual stage contributions are then multiplied by strength. Multiplied contributions are used to determine the winner in the individual stage. Given multiplied contributions of both agents, agent ig wins with the following probability:

$$p_{ig} = \begin{cases} \frac{\theta_{ig} y_{ig}}{\theta_{1g} y_{1g} + \theta_{2g} y_{2g}} & \text{if} \quad \theta_{1g} y_{1g} + \theta_{2g} y_{2g} > 0 \\ \frac{\theta_{ig}}{\theta_{1g} + \theta_{2g}} & \text{otherwise} \end{cases} \tag{2}$$

3.3 Payoffs

Each agent earns her endowment minus her group stage contribution. Members of the winning group also sacrifice individual stage contributions. Ultimately only one agent out of four earns the prize V. This agent belongs to the winning group and wins the individual stage. Given the above probabilities of winning in each stage, the expected payoff for the two-stage contest for agent ig is:

$$\pi_{ig} = w - x_{ig} + p_g(p_{ig}V - y_{ig}) \tag{3}$$

3.4 Strength

The main innovation of the model is the freedom to vary the endogeneity and asymmetry of strengths in the individual stage. Through the strength parameter, I establish a novel link between the group stage and the individual stage. In the individual stage, an agent's strength is a function of her group stage contribution. In general, $\theta_{ig} = f(x_{ig})$ where f represents the *strength function*. As a baseline, I consider when strength is exogenous and symmetric across all subjects such that $\theta_{ig} = c \quad \forall \quad ig$. In this case, regardless of group stage decisions, strength is known, constant and equivalent across all agents.

I also consider two cases in which strength is endogenous to group stage contribution. In one case f is monotonically increasing and concave in x_{ig}. Higher group stage contributions translate into higher strength. In another case, f is monotonically decreasing and concave in x_{ig}. In this case, the more an agent contributes to her group, the lower her strength in the individual stage.

I also consider an additional case in which strength is exogenous and asymmetric. For each subject, a strength is drawn randomly from a uniform distribution on the interval [1,5]. In the analysis that follows, I discuss equilibrium and comparative statics for these cases.

3.5 Analysis

I now characterize subgame-perfect equilibria of the two-player, two-stage contest with potentially endogenous strength. The model is solved via backward induction. Without loss of generality, fix the winning group in the individual stage to be group a and define

$$\theta_{ia} = f(x_{ia}), \qquad r_{ia} = w - x_{ia}.$$

where r_{ia} denotes the remaining endowment. Each agent $ia \in \{1a, 2a\}$ chooses $y_{ia} \in [0, r_{ia}]$ to maximize

$$u_{ia}(y_{ia}, y_{ja}) = \frac{\theta_{ia}y_{ia}}{\theta_{1a}y_{1a} + \theta_{2a}y_{2a}}V - y_{ia} + r_{ia}.$$

Lemma 1 (Individual stage Nash equilibrium)**.** *Assuming non-binding budget constraints*[3]*, given any vector of strengths* $(\theta_{1a}, \theta_{2a})$*, the individual stage has a unique symmetric Nash equilibrium in which*

$$y^*_{1a} = y^*_{2a} = \frac{V\theta_{1a}\theta_{2a}}{(\theta_{1a} + \theta_{2a})^2}.$$

[3]for a discussion of the individual stage Nash equilibrium when budget constraints bind, see Appendix A.1.

Winning probabilities are

$$p_{1a}^* = \frac{\theta_{1a}}{\theta_{1a} + \theta_{2a}}, \qquad p_{2a}^* = \frac{\theta_{2a}}{\theta_{1a} + \theta_{2a}},$$

and continuation payoffs are

$$U_{1a} = \frac{V\theta_{1a}^2}{(\theta_{1a} + \theta_{2a})^2}, \qquad U_{2a} = \frac{V\theta_{2a}^2}{(\theta_{1a} + \theta_{2a})^2}.$$

Proof. Assume for all (x_{1a}, x_{2a}) that arise in equilibrium, budget constraints do not bind in the individual stage:

$$w - x_{ia} \;\geq\; \frac{V\theta_{1a}\theta_{2a}}{(\theta_{1a} + \theta_{2a})^2} \quad \text{for } ia \in \{1a, 2a\},$$

and $f(x) > 0$ on $[0, w]$. The payoff function is strictly concave in y_{ia} for $y_{ja} > 0$, so first-order conditions are sufficient and yield the unique interior Nash equilibrium. The FOCs are

$$\frac{V\theta_{1a}\theta_{2a}y_{2a}}{(\theta_{1a}y_{1a} + \theta_{2a}y_{2a})^2} - 1 = 0 \qquad \frac{V\theta_{1a}\theta_{2a}y_{1a}}{(\theta_{1a}y_{1a} + \theta_{2a}y_{2a})^2} - 1 = 0$$

which implies $y_{1a}^* = y_{2a}^*$. Substitution gives the winning probabilities and continuation payoffs. □

Notice that the individual stage Nash equilibrium is symmetric even if agents are asymmetric, but equilibrium effective contributions (contributions multiplied by strength), probabilities of winning, and expected payoffs differ between the strong and weak agent.

Figure 1 illustrates how the individual stage Nash equilibrium changes with strengths using example parameters that match the parameters I use in the experiment. The horizontal axis is the ratio of strengths. A strength ratio of 1 means perfect symmetry between agents. Notice that the stage Nash equilibrium is maximized whenever agents are symmetric and decreases as agents become more asymmetric. The intuition is that higher inequality between agents dampens competition and puts downward pressure on contributions. The strong agent can achieve the same probability of winning with a lower contribution, so he finds it optimal to decrease contributions relative to the equilibrium when both agents are equally strong. The weak agent finds it optimal to decrease contributions as well. The greater the wedge between the strong and weak agent, the lower the equilibrium contribution level.

Importantly, strengths should be taken as given by both agents in the individual stage, and the equilibrium remains identical regardless of how strengths were determined. Therefore, whether strengths are randomly assigned or endogenously earned by a prior decision should not affect the individual stage equilibrium as long as the budget constraint doesn't bind.

Remark 1 (Strength assignment)**.** *The individual stage Nash equilibrium depends only on the vector of strengths and not on how strengths are assigned.*

Moving backwards to the group stage and substituting individual stage continuation values conditional on winning, the expected payoff of agent ig in the group stage is

$$\pi_{ig}(x_{ig}, x_{jg}, X_{-g}) = w - x_{ig} + p_g(X_g, X_{-g})\, U_{ig}\big(f(x_{ig}), f(x_{jg})\big) \tag{4}$$

Agent ig chooses $x_{ig} \in [0, w]$ to maximize expected two-stage payoff. I first consider symmetric equilibrium group stage actions.

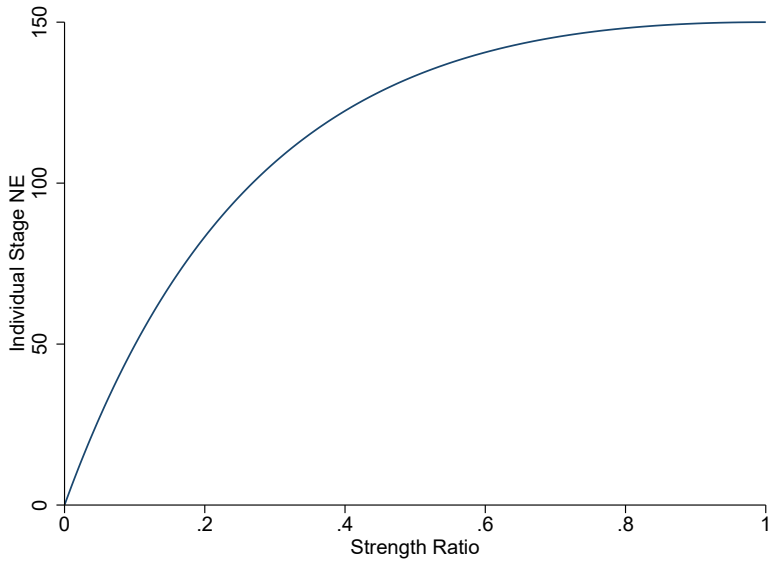


Figure 1: Individual Stage NE by Strength Ratio, Experimental Parameters

Lemma 2 (Symmetric interior equilibrium condition)**.** *Any symmetric interior group stage equilibrium $x^* > 0$ satisfies*

$$1 = \frac{V}{32x^*} + \frac{V}{8}\frac{f'(x^*)}{f(x^*)}.$$

Proof. Consider agent $1a$. Differentiate π_{1a} with respect to x_{1a} and impose the first-order condition. The simplified FOC results in

$$2\frac{X_a}{X_a + X_b}\left(\frac{f(x_{1a})f(x_{2a})f'(x_{1a})}{(f(x_{1a}) + f(x_{2a}))^3}\right) + \frac{X_b}{(X_a + X_b)^2}\frac{f(x_{1a})^2}{(f(x_{1a}) + f(x_{2a}))^2} = 1 \tag{5}$$

A symmetric FOC exists for each agent. Consider a symmetric profile

$$x_{1a} = x_{2a} = x_{1b} = x_{2b} = x.$$

Then $X_a = X_b = 2x$ and $f(x_{1a}) = f(x_{2a}) = f(x_{1b}) = f(x_{2b}) = f(x)$. Within the winning group,

$$U_{1a} = U_{2a} = \frac{V}{4}.$$

Imposing symmetry and simplifying the above FOC further results in the symmetric equilibrium condition. □

This condition highlights the two forces governing group stage investment. The first term reflects the effect of contributions on the probability of advancing to the individual stage, while the second term captures the effect of contributions on strength and thus the continuation value in the individual stage.

3.5.1 Exogenous Symmetric Strength

As a benchmark, suppose strength is exogenous and symmetric across all agents:

$$\theta_{ig} = \theta > 0 \qquad \forall\, i, g.$$

Then the individual stage (conditional on reaching it) is a standard symmetric two-player Tullock contest. By Lemma 1, the unique (interior) Nash equilibrium of the individual stage satisfies

$$y^*_{1g} = y^*_{2g} = \frac{V}{4}, \qquad p^*_{1g} = p^*_{2g} = \frac{1}{2}, \qquad u_{1g} = u_{2g} = \frac{V}{4},$$

where $u_{ig} = p^*_{ig}V - y^*_{ig}$ denotes the continuation payoff conditional on the group winning the group stage (net of the constant $r_{ig} = w - x_{ig}$ term).

Substituting $u_{ig} = V/4$ into the group stage payoff, agent ig's expected two-stage payoff in the group stage becomes

$$\pi_{ig}(x_{ig}, x_{jg}, X_{-g}) = p_g(X_g, X_{-g})\frac{V}{4} - x_{ig} + w, \tag{6}$$

where X_{-g} is the other group's total contributions.

Proposition 1 (Multiple subgame-perfect equilibria with exogenous strength)**.** *Assume $\theta_{ig} = \theta > 0$ for all ig and that individual stage budget constraints do not bind. Then there is a continuum of subgame-perfect equilibria. In any such equilibrium,*

$$X^*_a = X^*_b = \frac{V}{16},$$

and within each group $g \in \{a, b\}$ any split of the total group contribution is consistent with equilibrium:

$$x^*_{1g} \in \left[0, \frac{V}{16}\right], \qquad x^*_{2g} = \frac{V}{16} - x^*_{1g}.$$

In the individual stage, the unique equilibrium is $y^*_{1g} = y^*_{2g} = V/4$.

Proof. Because the individual stage equilibrium is unique and yields $u_{ig} = V/4$ for both agents in the winning group, stage-1 equilibrium reduces to Nash equilibrium of the group stage game with payoffs equal to (6).

Differentiating expected payoff with respect to x_{ig} yields

$$\frac{\partial \pi_{ig}}{\partial x_{ig}} = -1 + \frac{V}{4} \cdot \frac{\partial}{\partial x_{ig}} \left(\frac{x_{ig} + x_{jg}}{x_{ig} + x_{jg} + X_{-g}} \right) = -1 + \frac{V}{4} \cdot \frac{X_{-g}}{(X_g + X_{-g})^2}.$$

Since

$$\frac{\partial^2 \pi_{ig}}{\partial x_{ig}^2} = \frac{V}{4} \cdot X_{-g} \cdot \frac{\partial}{\partial x_{ig}} (X_g + X_{-g})^{-2} = -\frac{V}{2} \cdot \frac{X_{-g}}{(X_g + X_{-g})^3} < 0 \quad \text{whenever } X_{-g} > 0,$$

the objective is strictly concave in x_{ig} for $X_{-g} > 0$, so the first-order condition is sufficient for a unique best response.

An interior best response satisfies the following FOCs:

$$-1 + \frac{V}{4} \cdot \frac{X_b}{(X_a + X_b)^2} = 0 \qquad -1 + \frac{V}{4} \cdot \frac{X_a}{(X_a + X_b)^2} = 0$$

which implies $X_a = X_b$. An equilibrium can only pin down aggregate group stage contributions. Within each group g, any split (x_{1g}, x_{2g}) with $x_{1g}, x_{2g} \geq 0$ and $x_{1g} + x_{2g} = V/16$ constitutes mutual best responses given that the opposing group's total is also $V/16$.

The individual stage equilibrium follows from Lemma 1 and is unique. □

3.5.2 Endogenous Strength

In the general model, the strength function f is unrestricted. However, without additional structure, the reduced group stage game may admit multiple equilibria, including asymmetric equilibria, which complicates both theoretical characterization and empirical predictions. To obtain sharp and tractable predictions, I restrict attention to the following class of strength functions:

$$f(x) = (x + \alpha)^{\gamma}, \qquad \alpha > 0,\ \gamma \in (-1, 1) \setminus \{0\}.$$

This class of functions is strictly positive and continuously differentiable on $[0, w]$, ensuring well-defined contest success functions and facilitating backward induction. When $\gamma \in (0, 1)$, strength is increasing and exhibits diminishing returns, while when $\gamma \in (-1, 0)$, strength is decreasing in group stage contributions.

These strength functions permit tractable equilibrium characterization, allowing for closed-form solutions and transparent comparative statics. The experimental treatments correspond to two particular cases within this class: $f(x) = (x + 3)^{1/3}$, which features positive monotonically increasing concave strength, and $f(x) = (x + 3)^{-1/3}$, which features positive monotonically decreasing convex strength. These cases generate distinct incentives linking the group stage to the individual stage while remaining analytically tractable. The bounded curvature implied by $\gamma \in (-1, 1)$ ensures well-behaved equilibrium behavior and, in the cases studied in the experiment, yields a unique equilibrium.

Proposition 2 (Unique symmetric subgame-perfect equilibrium with endogenous strength)**.** *Assume individual stage budget constraints do not bind and let*

$$f(x) = (x + \alpha)^{\gamma}, \qquad \alpha > 0,\ \gamma \in (-1, 1) \setminus \{0\}.$$

Then the two-stage game admits a unique symmetric subgame-perfect equilibrium. In this equilibrium, group stage contributions are symmetric and equal across all agents,

$$x^*_{1a} = x^*_{2a} = x^*_{1b} = x^*_{2b} = x^* > 0,$$

The symmetric group stage contribution x^* is the unique solution in $(0, w]$ to

$$1 = \frac{V}{32x} + \frac{V}{8}\frac{f'(x)}{f(x)} = \frac{V}{32x} + \frac{V\gamma}{8(x + \alpha)}, \tag{7}$$

and is given in closed form by

$$x^* = \frac{V(1 + 4\gamma) - 32\alpha + \sqrt{(32\alpha - V(1 + 4\gamma))^2 + 128V\alpha}}{64}. \tag{8}$$

Individual stage contributions are given by the unique Nash equilibrium characterized in Lemma 1 for any vector of strengths.

Proof. By Lemma 1, the individual stage has a unique Nash equilibrium in which

$$y^*_{1g} = y^*_{2g} = \frac{V\theta_{1g}\theta_{2g}}{(\theta_{1g} + \theta_{2g})^2}, \qquad p^*_{ig} = \frac{\theta_{ig}}{\theta_{1g} + \theta_{2g}}, \qquad u_{ig} = \frac{V\theta^2_{ig}}{(\theta_{1g} + \theta_{2g})^2}.$$

In particular, under a symmetric group stage profile, $\theta_{1g} = \theta_{2g} = f(x)$ so $y^*_{1g} = y^*_{2g} = V/4$ and $u_{1g} = u_{2g} = V/4$.

Substituting the continuation values into the group stage payoff yields

$$\pi_{ig}(x_{ig}, x_{jg}, X_{-g}) = w - x_{ig} + p_g(X_g, X_{-g})\, U_i\big(f(x_{ig}), f(x_{jg})\big).$$

By Lemma 2, any symmetric interior equilibrium $x^* > 0$ must satisfy

$$1 = \frac{V}{32x^*} + \frac{V}{8}\frac{f'(x^*)}{f(x^*)}.$$

Under $f(x) = (x+\alpha)^\gamma$, this condition becomes the equality (7).

Define the symmetric first-order condition

$$FOC(x) \equiv \frac{V}{32x} + \frac{V\gamma}{8(x+\alpha)} - 1, \qquad x \in (0, w].$$

Then FOC is continuous on $(0, w]$ and $\lim_{x\downarrow 0} FOC(x) = +\infty$. Moreover,

$$FOC'(x) = -\frac{V}{32x^2} - \frac{V\gamma}{8(x+\alpha)^2} < 0,$$

so FOC is strictly decreasing. Hence $FOC(x) = 0$ has at most one solution, and if $FOC(w) \le 0$, existence and uniqueness of $x^* \in (0, w]$ follow by the Intermediate Value Theorem.

Individual stage equilibrium is uniquely pinned down by Lemma 1 for every realized vector of strengths. Uniqueness holds because the symmetric group stage equilibrium is unique and the individual stage equilibrium is unique given strengths. □

Corollary 1 (No asymmetric group stage equilibrium in special cases)**.** *Assume individual stage budget constraints do not bind and let*

$$f(x) = (x+3)^\gamma, \qquad \gamma \in \left\{\tfrac{1}{3}, -\tfrac{1}{3}\right\}.$$

Then a subgame-perfect equilibrium does not admit asymmetric group stage contributions. In particular, in any equilibrium,

$$x^*_{1a} = x^*_{2a} = x^*_{1b} = x^*_{2b} = x^* > 0,$$

Proof. By Lemma 1, conditional on group g winning the group stage, agent ig's continuation payoff is

$$u_{ig}(x_{ig}, x_{jg}) = V\frac{f(x_{ig})^2}{\big(f(x_{ig}) + f(x_{jg})\big)^2} - x_{ig} + w. \tag{9}$$

So the reduced two-stage payoff is

$$\pi_{ig} = w - x_{ig} + p_g\, u_{ig}(x_{ig}, x_{jg}).$$

The first-order condition for agent ig is

$$-1 + p'_g u_{ig}(x_{ig}, x_{jg}) + p_g \frac{\partial u_{ig}}{\partial x_{ig}}(x_{ig}, x_{jg}) = 0. \tag{10}$$

Towards a contradiction, suppose that there exists an interior equilibrium with $x_{1g} \neq x_{2g}$. Without loss of generality, let $x_{1g} > x_{2g}$. Subtracting the FOCs in (10) for agents $1g$ and $2g$ yields

$$p_g'(u_{1g} - u_{2g}) + p_g\left(\frac{\partial u_{1g}}{\partial x_{1g}} - \frac{\partial u_{2g}}{\partial x_{2g}}\right) = 0. \tag{11}$$

Differentiating equation (9) gives

$$\frac{\partial u_{ig}}{\partial x_{ig}} = 2V \cdot \frac{f(x_{ig})f(x_{jg})}{\big(f(x_{ig}) + f(x_{jg})\big)^3} f'(x_{ig}). \tag{12}$$

Substituting into equation (11) and simplifying yields

$$\frac{p_g'}{p_g} = -2 \cdot \frac{f(x_{1g})f(x_{2g})}{\big(f(x_{1g}) + f(x_{2g})\big)^2} \cdot \frac{f'(x_{1g})/f(x_{1g}) - f'(x_{2g})/f(x_{2g})}{\frac{f(x_{1g})^2 - f(x_{2g})^2}{f(x_{1g})f(x_{2g})}}. \tag{13}$$

First note that the left-hand side of equation (13) is strictly positive:

$$\frac{p_g'}{p_g} = \frac{X_{-g}}{X_g(X_g + X_{-g})} > 0.$$

Now consider the right-hand side. When $x_{1g} > x_{2g}$ and $\gamma \in \{\frac{1}{3}, -\frac{1}{3}\}$, we have that f is strictly monotone and $f'(x)/f(x) = \gamma/(x+3)$. Hence

$$\frac{f'(x_{1g})}{f(x_{1g})} - \frac{f'(x_{2g})}{f(x_{2g})} = \frac{\gamma}{x_{1g}+3} - \frac{\gamma}{x_{2g}+3},$$

which has the opposite sign of γ. Moreover,

$$f(x_{1g})^2 - f(x_{2g})^2$$

has the same sign as γ. It follows that the fraction in equation (13) is negative when $\gamma > 0$ and positive when $\gamma < 0$, implying that the entire right-hand side of equation (13) is strictly negative for both $\gamma = \pm 1/3$:

$$-2 \cdot \frac{f(x_{1g})f(x_{2g})}{\big(f(x_{1g}) + f(x_{2g})\big)^2} \cdot \frac{f'(x_{1g})/f(x_{1g}) - f'(x_{2g})/f(x_{2g})}{\frac{f(x_{1g})^2 - f(x_{2g})^2}{f(x_{1g})f(x_{2g})}} < 0.$$

This contradicts equation (13), since its left-hand side is positive. Therefore, no interior equilibrium can satisfy $x_{1g} > x_{2g}$. By symmetry of the argument, $x_{2g} > x_{1g}$ is also impossible. Hence $x_{1g} = x_{2g}$ in any interior equilibrium.

Since groups are symmetric, it must be that $X_a = X_b$ in equilibrium. If $X_a = X_b$ and $x_{1g} = x_{2g}$ for all g, then it must be that $x_{1a} = x_{2a} = x_{1b} = x_{2b} = x$. Therefore, any equilibrium must have symmetric group stage contributions. □

Corollary 2 (Comparative statics across experimental treatments)**.** *Assume budget constraints do not bind. Consider the following treatments:*

1. ***Symmetric strength (SYM):*** $f(x) = \theta > 0$.

2. ***Increasing strength (INC):*** $f(x) = (x+3)^{1/3}$.

3. ***Decreasing strength (DEC):*** $f(x) = (x+3)^{-1/3}$.

Let x_T^* denote the symmetric equilibrium group stage contribution of each agent under treatment $T \in \{INC, SYM, DEC\}$, and let y_T^* denote the symmetric equilibrium individual stage contribution of each agent. Then:

1. ***Group stage contributions.***

$$x_{INC}^* > x_{SYM}^* > x_{DEC}^*.$$

2. ***Individual stage contributions.***

$$y_{INC}^* = y_{SYM}^* = y_{DEC}^*.$$

3. ***Total expenditure.*** *Fix a as the winning group and let total expenditure be defined as*

$$TOT_T^* = \frac{X_a + y_{1a} + y_{2a}}{V}.$$

Then

$$TOT_{INC}^* > TOT_{SYM}^* > TOT_{DEC}^*.$$

4. ***Expected equilibrium payoffs.*** *Let $\mathbb{E}[\pi^*]_T$ denote expected equilibrium payoff in treatment T. Then*

$$\mathbb{E}[\pi^*]_{INC} < \mathbb{E}[\pi^*]_{SYM} < \mathbb{E}[\pi^*]_{DEC}.$$

Proof. Follows directly from Proposition 1 and Proposition 2. □

3.5.3 Exogenous Asymmetric Strength

Finally, I consider the case in which strength is exogenous but asymmetric. Suppose each agent's strength θ_{ig} is independently drawn from a continuous uniform distribution on $[1, 5]$. In the group stage, each agent observes only her own strength and knows the distribution from which all strengths are drawn. An agent does not observe the strength of her group mate. Once a group advances to the individual stage, both members observe realized strengths. Lemma 1 characterizes the unique equilibrium in the individual stage and the corresponding continuation payoffs. Conditional on group g winning the group stage, agent ig's continuation payoff is

$$U_{ig} = \frac{V\theta_{ig}^2}{(\theta_{ig} + \theta_{jg})^2}. \tag{14}$$

Differentiating with respect to own strength yields

$$\frac{\partial U_{ig}}{\partial \theta_{ig}} = \frac{2V\theta_{ig}\theta_{jg}}{(\theta_{ig} + \theta_{jg})^3} > 0, \tag{15}$$

which indicates that continuation payoffs in the individual stage increase in own strength.

Moving backwards to the group stage, an agent observes her own strength but not the strength of any other player. Let $x(\theta_{ig})$ denote a Bayesian equilibrium contribution strategy,

where $x(\theta_{ig})$ specifies the group stage contribution of an agent with strength θ_{ig}. Consider agent $1a$ who observes θ_{1a} and chooses $x(\theta_{1a}) \in [0, w]$ to maximize

$$\pi_{1a} = w - x(\theta_{1a}) + \mathbb{E}\left[\frac{x(\theta_{1a}) + x(\theta_{2a})}{x(\theta_{1a}) + x(\theta_{2a}) + x(\theta_{1b}) + x(\theta_{2b})} \frac{\theta_{1a}^2}{(\theta_{1a} + \theta_{2a})^2}\right], \tag{16}$$

where the expectation is taken over all possible draws of the unobserved strengths. Differentiating expected payoff with respect to $x(\theta_{1a})$ yields

$$1 = \mathbb{E}\left[\frac{x(\theta_{1b}) + x(\theta_{2b})}{(x(\theta_{1a}) + x(\theta_{2a}) + x(\theta_{1b}) + x(\theta_{2b}))^2} \frac{\theta_{1a}^2}{(\theta_{1a} + \theta_{2a})^2}\right]. \tag{17}$$

where the right hand side represents the marginal benefit of increasing group stage contribution.

I do not offer a closed-form characterization of the equilibrium contribution function $x(\theta_{ig})$ which specifies the optimal group stage contribution for every strength. Instead, I make one remark about the relationship between observed own strength and equilibrium group stage contribution to guide predictions about behavior in the experiment. Since continuation value increases in own strength, agents who draw a higher strength face a higher expected reward from winning the group stage and advancing to the individual stage. Consequently, equilibrium group stage contribution increases in observed own strength.

Remark 2 (Group stage contributions and strength)**.** *When strength is randomly drawn for each agent from a uniform distribution, group stage contributions are strictly increasing in strength. That is, for* $\theta', \theta \sim U[1, 5]$ *with* $\theta' > \theta$*,*

$$x(\theta') > x(\theta). \tag{18}$$

Proof. Consider two strengths $\theta' > \theta$. Suppose, toward a contradiction, that

$$x(\theta') \leq x(\theta). \tag{19}$$

In equilibrium, given realized strength, group stage contribution satisfies the first order condition in (17). Therefore, for type θ' and θ, the marginal benefit of increasing group stage contribution equals 1, $MB(\theta') = 1$ and $MB(\theta) = 1$. The marginal benefit strictly increases in strength and decreases in group stage contribution. Therefore, if $\theta' > \theta$ and $x(\theta') \leq x(\theta)$, then $MB(\theta') > MB(\theta) = 1$. But in equilibrium, $MB(\theta') = 1$, so we have a contradiction.

□

Essentially, higher strength translates to higher individual stage continuation value and higher marginal benefit of increasing group stage contribution. Therefore, group stage contribution increases in strength.

4 Experimental Design

I design and conduct a laboratory experiment in which subjects play the two-stage contest outlined above. I employ a between-subjects design across four treatments where I vary the relationship between group stage contribution and individual strength. In the baseline **Symmetric** treatment, strength is exogenously given and symmetric. In the **Increasing** treatment,

strength increases in group stage contribution. In the **Decreasing** treatment, strength decreases in group stage contribution. Finally, I run an additional **Random** treatment in which strength is exogenously given and randomly drawn for each subject from a uniform distribution on the interval [1,5]. Table 1 summarizes the experimental sessions, treatments and total participants. I run a total of 12 sessions consisting of 16 subjects per session for a total of 192 subjects (55% female). All experimental sessions take place at the XS/FS Lab at Florida State University and last at most 90 minutes. Each subject participates in exactly one session with no prior experience. I implement the experiment using z-Tree (Fischbacher, 2007) and recruit subjects in ORSEE (Greiner, 2015). Subjects earn $19.86 on average including a $7 show-up fee.

Table 1: Summary of Experimental Treatments

Treatment	**Sessions**	**Participants**
Increasing	3	48
Decreasing	3	48
Symmetric	3	48
Random	3	48
	12	192

In each session, subjects first watch a video recording of the experimenter reading over the experiment instructions.[4] Each subject also receives a paper copy of the instructions. Before proceeding to the decision rounds, subjects then complete a comprehension check quiz and participate in an unpaid 3-minute practice module that gives subjects practice with the experiment interface. Next, subjects complete the main incentivized part of the experiment involving a sequence of decision rounds. Finally, subjects complete a questionnaire consisting of unincentivized and optional questions. I elicit gender, risk preferences, and open responses on how subjects made their decisions.

In the main part of the experiment, subjects play a group contest followed by an individual contest. I create match groups of 8 subjects each and explain that subjects will only interact with members of their match group. Subjects play 20 repeated rounds of the two-stage contest. I randomly re-pair group mates and rematch groups within match groups between rounds. Each decision round consists of two stages. At the beginning of each round, each participant receives a one-time endowment of 300 points which they may contribute over both stages in the round.

The first stage is the group stage. In the group stage, two groups of two subjects each participate in a group contest. Groups compete only for the right to earn a prize in the next stage. Each individual chooses a group stage contribution from their endowment, and a group's contribution is the sum of the group stage contributions of both members. Whatever a subject decides not to contribute in this stage remains with her for the next stage.

Subjects in each treatment understand whether and how group stage contributions affect strength. In **Increasing** and **Decreasing**, subjects are given a calculator to determine the strength that will result from a group stage contribution before making a final submission. Once a group stage contribution is submitted, subjects immediately see their own strengths. In **Random**, subjects understand that each subject's strength is drawn from a uniform distribution

[4] A copy of the instructions for **Increasing** can be found in Appendix C. Instructions for the other three treatments are available upon request.

on the interval [1,5]. Each subject sees only her own strength on screen at the beginning of the group stage. However, subjects do not observe group mate strengths until the individual stage.[5]

One group wins the contest according to the group contest-success function described in the model above. However, all subjects advance to the next stage without learning the outcome of the group stage. Subjects see their contributions and their group mate's contributions as well as the corresponding remaining endowments. At this time, in **Increasing**, **Decreasing**, and **Random** each subject also learns the strength of her group mate.

The second stage is the individual stage. In the individual stage, subjects participate in the individual contest described in the model. The two group mates from the previous stage compete for a prize of 600 points. Subjects at this point don't know the outcome of the group stage and compete as if their group won. Each subject selects an individual stage contribution from her remaining endowment. Individual stage contributions are multiplied by strength, and multiplied contributions determine each subject's probability of winning the individual stage.

After subjects submit individual stage contributions, they see their group mate's decision and corresponding multiplied contribution and then learn the outcome of the round. At this point, I reveal the results of the group stage and the individual stage. Subjects who lose in the group stage earn their endowment minus their group stage contribution only. Subjects who win in the group stage but lose in the individual stage earn their endowment minus their group stage contribution and individual stage contribution. Finally, subjects who win in both stages earn an additional 600 points. After the last decision round, I randomly select 4 out of 20 rounds for payment. At the end of the experiment, I convert earnings in points to USD for payment at a rate of 1 point equals \$0.01.

I list testable hypotheses below which follow directly from the theoretical analysis in the previous sections. Since multiple equilibria exist in **Symmetric**, I focus on the average equilibrium group stage contribution for predictions and comparisons. The prediction can also be interpreted as the symmetric equilibrium prediction.

Hypothesis 1 (Group Stage)**.** *Group stage contributions are highest in* ***Increasing****, next highest in* ***Symmetric*** *and lowest in* ***Decreasing****.* $x^*_{INC} > x^*_{SYM} > x^*_{DEC}$*. In* ***Random****, group stage contributions are increasing in own strength.*

Hypothesis 2 (Individual Stage)**.** *Individual stage contributions are equivalent across all treatments.* $y^*_{INC} = y^*_{SYM} = y^*_{DEC}$*.*

Hypothesis 3 (Total Expenditure)**.** *Total expenditure is highest in* ***Increasing****, next highest in* ***Symmetric*** *and lowest in* ***Decreasing****.* $TOT^*_{INC} > TOT^*_{SYM} > TOT^*_{DEC}$*.*

Hypothesis 4 (Expected Payoffs)**.** *Expected payoffs are highest in* ***Decreasing****, next highest in* ***Symmetric*** *and lowest in* ***Increasing****.* $E[\pi^*]_{INC} < E[\pi^*]_{SYM} < E[\pi^*]_{DEC}$*.*

It is possible for subjects in different treatments to face similar vectors of strength in the individual stage. In theory, subjects should only care about the strengths themselves and not how those strengths were determined. Whether strengths were exogenously assigned or endogenously determined in a particular way shouldn't affect decisions. Strengths should be taken as

[5] Revealing only own strength most closely aligns with the information structure of the endogenous strength treatments in which subjects effectively choose own strength at the time of making a group stage contribution but don't yet know group mate's choice and corresponding strength.

Table 2: Summary Statistics, Last 10 Rounds

	Group Stage		Individual Stage	
Treatment	Mean	Prediction	Mean	Prediction
Increasing	60.35	42.09	121.03	150.00
	[39.24, 86.10]		[101.99, 138.24]	
Decreasing	17.48	4.19	160.73	150.00
	[3.75, 37.94]		[123.16, 194.17]	
Symmetric	32.82	18.75	174.10	150.00
	[22.79, 41.90]		[152.12, 188.44]	
Random	52.04	27.55	125.86	143.25
	[35.67, 67.14]		[109.04, 152.97]	

Notes: Mean contributions by stage and treatment. Last 10 Rounds. 95% wild cluster bootstrap confidence intervals in brackets.

given once the individual stage arises. Therefore, I propose an additional null hypothesis that individual stage behavior won't vary across treatments for a given vector of strengths. In the following section, I test this hypothesis and consider behavioral explanations to explain when differences arise across treatments contradicting the theoretical prediction.

Hypothesis 5 (Strength Assignment)**.** *For a given vector of strengths, individual stage contributions should not vary across treatments.*

5 Results

I analyze aggregate results on group stage contributions and individual stage contributions from 192 subjects over 12 sessions and four treatments. Since I randomly rematch within match-groups each round, I conduct non-parametric tests on match-group averages as independent observations. I also employ linear regression analysis. Due to the small number of independent clusters, I report 95% wild cluster bootstrap confidence intervals on regression coefficients, and I conduct all statistical inference using p-values computed with the wild cluster bootstrap method. I focus my main analysis on the last 10 rounds of the data and include select results from all rounds in Appendix B.

5.1 Group Stage and Individual Stage Contributions

Table 2 reports summary statistics and theoretical predictions for contributions in each stage and each treatment over the last 10 rounds. Predictions in **Increasing**, **Decreasing**, and **Symmetric** are derived from subgame perfect equilibria. For **Symmetric**, multiple equilibria exist, but I focus on the symmetric equilibrium. The prediction for **Random** is a benchmark derived from the mean strength draws given the distribution of strength used in the experiment. Data from **Random** will primarily be used to study individual stage contributions in a later section.

In the group stage, contributions exceed the equilibrium predictions on average for each

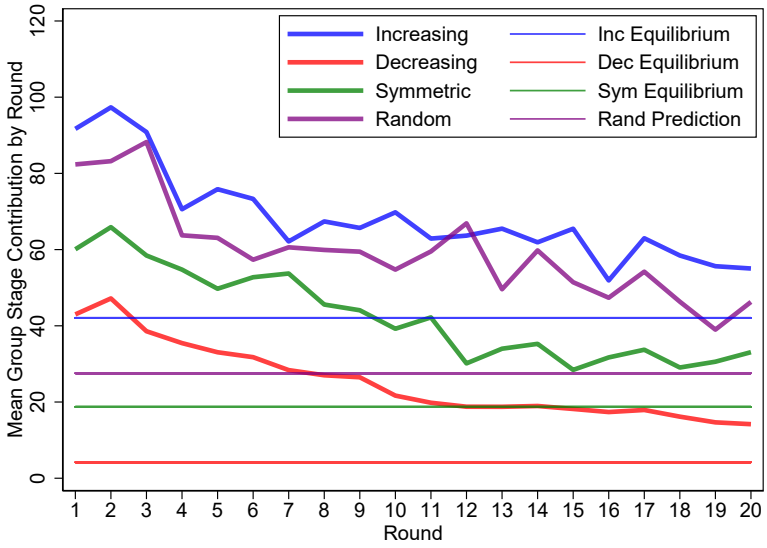


Figure 2: Group Stage Contributions by Round

treatment.[6] However, overinvestment diminishes over the course of 20 repeated rounds (see figure 2). As predicted by theory, group stage contributions in **Random** increase in own strength ($p = 0.03$; see table 11 in Appendix B).

In the individual stage, the subgame-perfect equilibrium in each treatment predicts a contribution of 150 points. Contrary to the typical result found in individual Tullock contest experiments, I observe little overinvestment in this stage relative to the subgame-perfect equilibrium benchmark. Focusing on the last 10 rounds, participants only significantly overinvest on average in **Symmetric**. Contributions in **Increasing** actually fall significantly below equilibrium prediction on average.[7].

Result 1 (Overinvestment relative to subgame-perfect equilibrium)**.** *Relative to subgame-perfect equilibrium predictions, subjects in all treatments overinvest in the group stage. In the individual stage, subjects do not overinvest on average in the endogenous strength treatments.*

5.1.1 Individual Stage Nash Equilibrium

Of course, a better benchmark for testing overinvestment in the individual stage is the realized Nash equilibrium for each group. Along the equilibrium path, group stage contributions are symmetric. Off equilibrium path group stage contributions may induce asymmetry in the individual stage which dampens competition and puts downward pressure on the unique symmetric individual stage Nash equilibrium (see figure 1).

I calculate the realized individual stage Nash equilibrium for each group of two subjects given their group stage contributions and corresponding strengths. Figure 4 plots the raw data

[6]Wald tests comparing constants in linear regression to predictions: INC, $p = 0.11$; DEC, $p = 0.06$; SYM, $p = 0.01$

[7]Wald tests comparing constants in linear regression to predictions: INC, $p = 0.01$; DEC, $p = 0.44$; SYM, $p = 0.04$

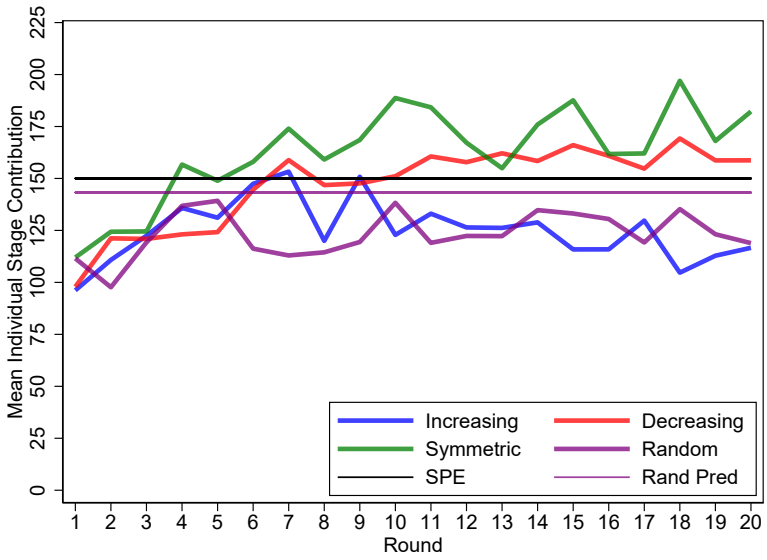


Figure 3: Individual Stage Contributions by Round

on the realized stage Nash equilibrium for each group. The horizontal axis shows the realized *strength ratio* for each individual stage contest. As above, the strength ratio is calculated as strength ratio $= \frac{\text{minimum strength}}{\text{maximum strength}}$ so that a higher strength ratio represents a more symmetric contest. Black Xs represent the NE given each observed strength ratio. Marks which fall off of the smooth curve represent cases in which budget constraints bind due to excessive group stage contributions. Even in these cases, I am able to calculate the asymmetric Nash equilibrium for the group. Colored dots then show individual stage contributions for each individual broken down by treatment.

I then calculate percent deviations from the stage Nash equilibrium. In figure 5, I plot mean percent deviations from stage Nash equilibrium by round for each treatment.[8] I find that the mean percent deviation in **Symmetric** is significantly greater than 0. The mean individual stage contribution in **Symmetric** over the last 10 rounds is 16.27% above the stage Nash equilibrium. The mean percent deviation from NE is actually negative for **Increasing** and **Random**. The mean individual stage contribution in **Increasing** over the last 10 rounds is 11.99% below the stage Nash equilibrium. This provides some suggestive evidence that when strength is increasing in group stage contributions, individual stage contributions fall not only below the subgame-perfect equilibrium prediction but also below the stage Nash equilibrium prediction. However, the difference from 0 percent deviation is not significant.[9] I also find that percent deviations are significantly lower in **Increasing** compared to the **Decreasing** and **Symmetric** but no

[8] I exclude contests in which at least one subject contributes the entire 300 point endowment in the group stage. In this case, the contestant who contributed everything must choose zero in the individual stage. The optimal best response for the other contestant is to contribute exactly one point. If this contestant submits something substantially larger than one, then the percent deviation is extreme and greatly affects the mean percent deviation calculations. Therefore, for this analysis, I drop cases in which subjects contribute everything in the group stage which only occurs twice in the last 10 rounds.

[9] Wald tests comparing constants in linear regressions to predictions: INC, $p = 0.14$; DEC, $p = 0.25$; SYM, $p = 0.04$; $RAND$, $p = 0.24$

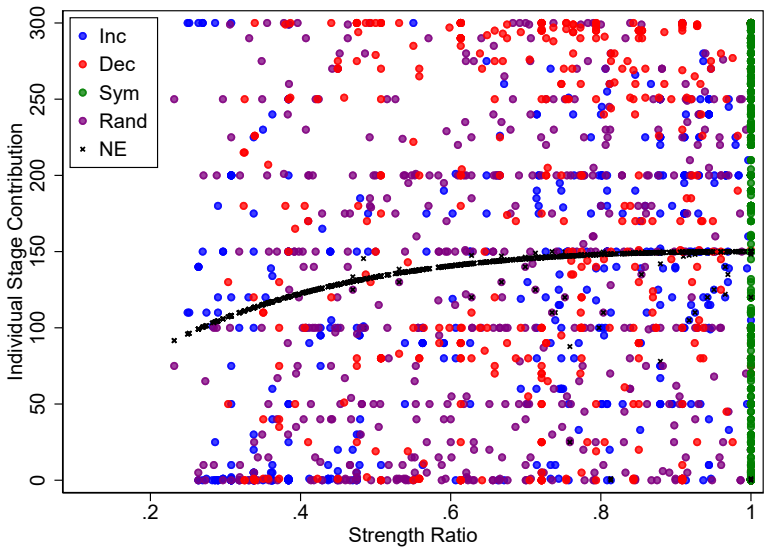


Figure 4: Individual Stage NE by Strength Ratio, Last 10 Rounds

different compared to **Random**.[10]

Many laboratory experiments involve subjects playing one-shot or repeated contests in isolation. Perhaps framing contests in dynamic environments that are more complex and realistic actually deters overinvestment.

Result 2 (Overinvestment relative to individual stage Nash equilibrium)**.** *Relative to individual stage Nash equilibrium predictions, subjects in the endogenous strength treatments do not significantly overinvest on average.*

Next, I compare contributions in each stage across treatments. Figures 2 and 3 depict mean contributions by round in each stage. Each figure also includes the theoretical predictions for each treatment. In both stages and in all treatments, substantial adjustments take place over the first 10 rounds after which behavior stabilizes considerably. In figures 12 and 14 in Appendix B, I also plot the empirical CDFs of contributions in each stage for each treatment for the last 10 rounds.

Figure 2 shows group stage contributions. Variations in group stage contributions across treatments follow the pattern predicted by the theory and in Hypothesis 1. According to pairwise Wilcoxon rank-sum tests on match-group averages, contributions are significantly higher in **Increasing** compared to each of the other treatments except **Random**. Contributions in **Symmetric** exceed contributions in **Decreasing**.[11] It appears subjects in each treatment anticipate the effect of group stage contributions on individual strength.

Result 3 (Group stage contributions)**.** *As the theory predicts group stage contributions are highest when the strength function is increasing and lowest when it is decreasing.*

[10] Wilcoxon rank-sum tests: INC vs. DEC, $p = 0.09$; INC vs. SYM, $p = 0.02$; INC vs. $RAND$, $p = 0.59$; DEC vs. SYM, $p = 0.94$; DEC vs. $RAND$, $p = 0.06$; SYM vs. $RAND$, $p = 0.009$

[11] Wilcoxon rank-sum tests: INC vs. DEC, $p = 0.009$; INC vs. SYM, $p = 0.04$; INC vs. $RAND$, $p = 0.59$; DEC vs. SYM, $p = 0.09$; DEC vs. $RAND$, $p = 0.009$; SYM vs. $RAND$, $p = 0.04$

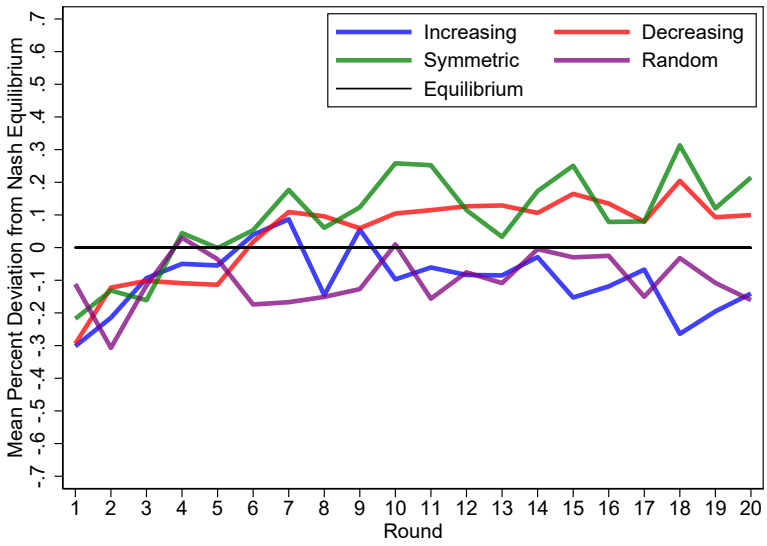


Figure 5: Percent Deviation from Individual Stage Equilibrium (excluding all-in groups)

Figure 3 shows mean individual stage contributions by round for each treatment. In the individual stage, group stage contributions are sunk. Subgame-perfect equilibria predict that subjects should arrive in the individual stage with equal strength regardless of treatment. Consequently, despite differences in group stage contribution levels, individual stage contributions should be the same across treatments. Contrary to the theoretical predictions and Hypothesis 2, I find significant differences on average across treatments which emerge and persist in the last 10 rounds. I find mean contributions in **Increasing** significantly below **Decreasing** and **Symmetric**. I find no significant difference between **Decreasing** and **Symmetric** or between **Increasing** and **Random**.[12]

Result 4 (Individual stage contributions)**.** *Contrary to theory, individual stage contributions are lower when the strength function is increasing versus when it is decreasing.*

The first two results analyze contributions in each stage in isolation. I now consider aggregate contributions over both stages before returning to the individual stage for additional analysis.

5.1.2 Total Contributions and Efficiency

So far, I observe significant differences between contributions in each stage in the endogenous strength treatments. Next, I consider aggregate contributions over both stages. I focus on expected payoffs and total expenditures over both stages within a group. These statistics serve as measures of market efficiency or rent dissipation. Figure 6 plots mean within-group total expenditures for each treatment and figure 7 plots mean expected payoffs. The figures also include equilibrium predictions and 95% wild cluster bootstrap confidence intervals around estimates.[13]

[12] Wilcoxon rank-sum tests: INC vs. DEC, $p = 0.06$; INC vs. SYM, $p = 0.02$; INC vs. $RAND$, $p = 0.70$; DEC vs. SYM, $p = 0.70$; DEC vs. $RAND$, $p = 0.06$; SYM vs. $RAND$, $p = 0.009$

[13] Tables showing mean estimates and confidence intervals are located in Appendix B

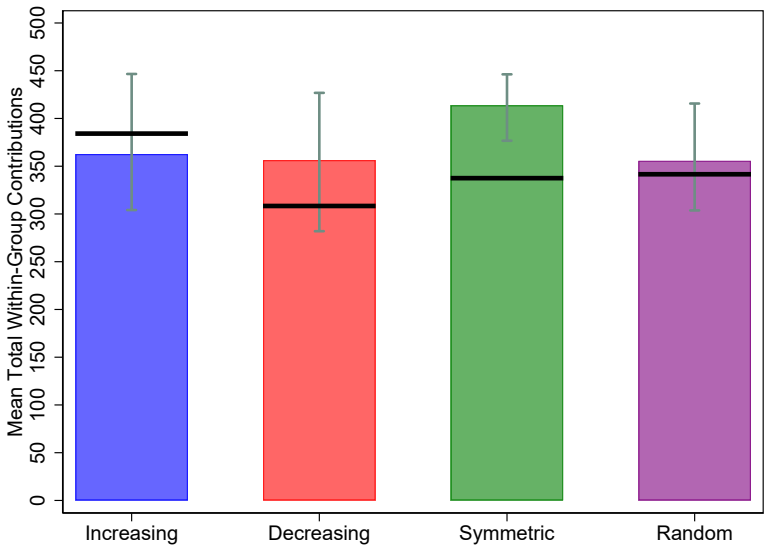


Figure 6: Within-Group Total Contributions, Last 10 Rounds

Theoretically, any differences across treatments in expected payoff or within group total expenditures should be solely driven by differences in group stage contribution as a consequence of varying the strength function. Empirically, endogenous strength affects contributions in both stages and the effects in each stage go in opposite directions. Ultimately, I observe no significant differences between the endogenous strength treatments in either measure of efficiency.

Regarding within-group total expenditures, pairwise Wilcoxon rank-sum tests using the last 10 rounds of the data show no significant differences between treatments except for when comparing **Symmetric** to **Random**.[14] In particular, contrary to the theory and Hypothesis 3, no significant difference emerges in expenditures between **Increasing** and **Decreasing**. Compared to the equilibrium predictions, within group total contributions are higher than expected in **Symmetric** and **Decreasing** but lower than expected in **Increasing** on average, and significantly lower in **Symmetric**.[15]

Next, contrary the theory and Hypothesis 4, I observe no significant difference in expected payoffs between the two endogenous strength treatments. I only observe a significant difference between **Decreasing** and **Symmetric** where **Decreasing** has a higher expected payoff than **Symmetric**.[16] Compared to the equilibrium predictions, expected payoffs are significantly lower on average in **Symmetric** and **Decreasing**.[17]

Result 5 (Total expenditures and expected payoffs)**.** *Total expenditures and expected payoffs do not differ significantly between treatments with endogenous strength. Varying the strength*

[14] Wilcoxon rank-sum tests: INC vs. DEC, $p = 0.94$; INC vs. SYM, $p = 0.24$; INC vs. $RAND$, $p = 1.00$; DEC vs. SYM, $p = 0.132$; DEC vs. $RAND$, $p = 0.82$; SYM vs. $RAND$, $p = 0.09$

[15] Wald tests comparing constants in linear regression to predictions: INC, $p = 0.46$; DEC, $p = 0.14$; SYM, $p = 0.008$

[16] Wilcoxon rank-sum tests: INC vs. DEC, $p = 0.31$; INC vs. SYM, $p = 0.59$; INC vs. $RAND$, $p = 1.00$; DEC vs. SYM, $p = 0.009$; DEC vs. $RAND$, $p = 0.31$; SYM vs. $RAND$, $p = 0.24$

[17] Wald tests comparing constants in linear regression to predictions: INC, $p = 0.51$; DEC, $p = 0.05$; SYM, $p = 0.005$

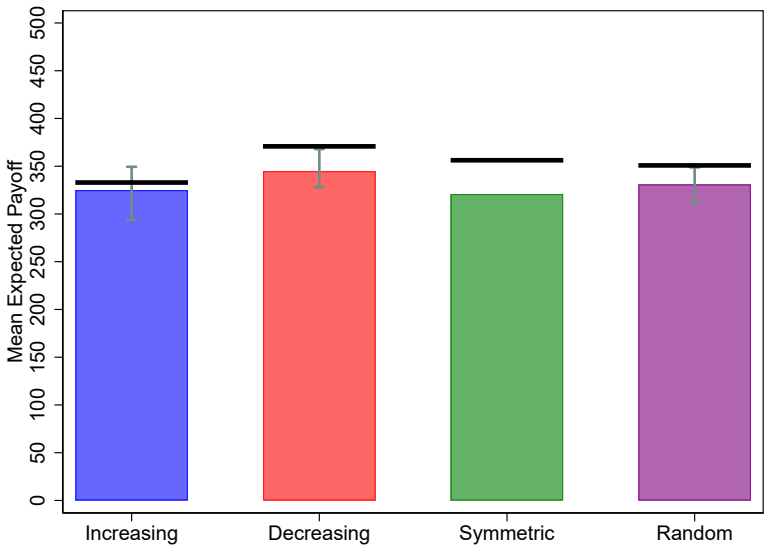


Figure 7: Expected Payoffs, Last 10 Rounds

function does not affect total contributions, but it does affect the stage in which contributions are expended.

In conjunction with Result 3 and Result 4, Result 5 is the first key contribution of the paper which offers potential implications for organizational and industrial policy. In the end, whether group stage contributions positively or negatively affect an individual's position in future internal conflict, total effort expenditure remains the same on average. However, the relationship between group effort and individual strength does substantially affect the timing and nature of effort expenditure. Specifically, when strength is increasing compared to when it is decreasing, group effort is higher and individual effort is lower.

If a manager or social planner cares only about aggregate effort expenditure, the incentive structure relating group competition and individual conflict does not matter. However, if the goal is to target and incentivize either group effort or individual effort and mitigate the other, the results from this experiment provide useful insight. For example, if group efforts are considered productive and individual efforts are considered wasteful, an increasing strength function would be preferred. By incentivizing higher group contributions through the promise of higher strength in internal competition, the principal actually discourages in-fighting without sacrificing collective effort. However, if only individual efforts are productive, then a decreasing strength function would be preferred to disincentivize group effort and heighten individual competition. Ultimately, the results of this paper suggest that an optimal organizational or industrial policy may depend on whether collective or individual effort is considered productive versus wasteful. Even if the relationship between contribution to group effort and strength is purely an artifact of a particular environment and not up to the discretion of the principal, then the results still provide descriptive information about how efforts are expended in different real-world settings.

For the next main set of results, I focus on the individual stage and explore how behavior varies across treatments with different relationships between the group stage and individual

stage.[18]

5.2 The Individual Stage

The individual stage in this experiment is a standard Tullock lottery between individuals. In **Increasing**, **Decreasing**, and **Random**, individuals may possess asymmetric strengths. This type of asymmetric contest closely resembles the design of Fonseca (2009) which compares symmetric and asymmetric contests. The main novelty in my design is that the contest is preceded by a group contest in which group mates previously worked together against another group. Furthermore, strengths are determined in the prior group stage either by a player's choice or by random draw. Recall Hypothesis 5 which predicts that only the strengths themselves should impact behavior in the individual stage. How strengths are assigned should not matter.

In the analyses that follow, I study how preceding an individual contest with a group contest and embedding a link between the two affects behavior. I compare contributions to equilibrium predictions and between treatments and test whether individuals respond differently to asymmetry depending on how asymmetry arises.[19]

5.2.1 Robustness Tests

I first test the robustness of the treatment differences in individual stage contributions to various regression specifications. Table 3 shows the results for six different specifications regressing individual stage contributions on treatment dummies and various control variables. In all specifications, **Random** is the excluded dummy variable.

The first specification is the naive model without any controls. Comparing mean individual stage contributions via regression analysis yields the same conclusions as the rank-sum tests.[20] The second and third specifications control for variables from the group contest. Group contribution is simply group stage contribution, and group probability of winning is a group's probability of winning the group contest which appears on screen after group stage contributions are made. Treatment differences are robust to these two specifications.

[18] In tables 16 and 17 in Appendix B, I analyze the effects of group stage and individual stage contributions on previous round history. I include lags for previous round contributions and outcomes. I find that group stage contributions increase significantly in last-round group total contributions for all treatments and in winning the prize only for **Increasing**. Individual stage contributions also increase significantly in last round contributions in all treatments and in winning the prize in **Increasing**. Individual stage contributions decrease if the group won in the last round but the individual didn't win the prize.

[19] Recall that in my design, I use a "strategy method" approach. I withhold the outcome of the group stage until the end of each round, so all groups participate in the individual stage. However, the individual stage only affects payoffs for the group that wins. After submitting group stage contributions, subjects do see the probability that their group wins. There is a possibility that this could influence behavior in the individual stage. In figure 15 in Appendix B, I plot a positive relationship between the sum of individual stage contributions and the group probability of winning for each treatment. Individuals internalize the likelihood that the individual stage will affect their round payoffs and behave more competitively. This is consistent and similar across all treatments. I rerun the results of the following section on only the groups that actually win, only the groups that have a probability of winning greater than 0.5, and only the groups that have a probability of winning greater than 0.33. While precision and hypothesis test results change occasionally compared to studying the full sample, most of the qualitative results remain the same albeit with a higher level of individual stage contributions across the board. Therefore, I utilize the full sample for the following analyses.

[20] Wald tests comparing constants in linear regression: INC vs DEC, $p = 0.03$; INC vs SYM, $p = 0.002$; INC vs. $RAND$, $p = 0.68$; DEC vs. SYM, $p = 0.43$; DEC vs. $RAND$, $p = 0.07$; SYM vs. $RAND$, $p = 0.005$

Table 3: Individual Stage Contributions and Controls, Last 10 Rounds

	(1)	(2)	(3)	(4)	(5)	(6)
Symmetric	48.246***	51.766***	48.246***			
	[23.755,72.737]	[25.662,77.869]	[23.749,72.743]			
Increasing	-4.829	-6.350	-4.829	-5.230	-7.187	-2.084
	[-27.091,17.432]	[-27.668,14.968]	[-27.097,17.438]	[-27.841,17.382]	[-30.584,16.209]	[-28.074,23.906]
Decreasing	34.871*	41.201*	34.871*	35.686*	27.509	29.159
	[2.631,67.111]	[8.045,74.357]	[2.623,67.119]	[2.081,69.291]	[-5.806,60.824]	[-5.333,63.650]
Group contribution		0.183				
		[-0.035,0.402]				
Group probability of winning			93.897***			
			[62.201,125.593]			
Relative Strength				10.090		
				[-5.050,25.230]		
Strength Ratio					73.466*	339.141*
					[17.582,129.349]	[0.320,677.963]
Strength Ratio Squared						-202.097
						[-454.248,50.054]
Constant	125.856***	116.325***	78.908***	113.853***	78.384***	-0.945
	[108.546,143.166]	[94.624,138.027]	[55.386,102.429]	[88.640,139.066]	[38.413,118.356]	[-109.243,107.353]
Observations	1920	1920	1920	1440	1440	1440
Clusters	24	24	24	18	18	18

Notes: Individual stage contribution on treatment and controls. Last 10 rounds. Wild cluster bootstrapping 95% confidence intervals in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The remaining specifications control for different measures of asymmetry. Relative strength is calculated for each subject as strength divided by group mate's strength. This measure of asymmetry captures both the degree of asymmetry in the contest as well as a particular subject's relative position. Strength ratio as defined above is calculated for each group as the minimum strength divided by the maximum strength. The strength ratio isolates the degree of asymmetry in the contest and ignores which player is stronger. The final specification includes a quadratic term for strength ratio given the non-linear theoretical relationship between strength ratio and individual stage Nash equilibrium. Since there is no strength variation in **Symmetric**, those observations are dropped for these models.

Results are fairly robust to controlling for each measure of asymmetry, except the difference between individual stage contributions in **Decreasing** and **Random** becomes insignificant in specifications 5 and 6.[21]

All in all, differences in individual stage contributions across treatments remain robust to controlling for different measures of group stage behavior or individual stage asymmetry. This warrants further investigation into what drives the treatment differences we observe.

5.2.2 Effect of Group Stage Contributions

I first consider the relationship between group stage contributions and individual stage contributions for each treatment. In table 4, I regress individual stage contributions on group stage contributions interacted with treatment dummies. Individual stage contributions increase

[21] Wald tests comparing constants in linear regression: DEC vs. $RAND$, $p = 0.13$ in specification 5; DEC vs. $RAND$, $p = 0.13$ in specification 6

Table 4: Individual Stage Contributions, Last 10 Rounds

	(1)
Increasing	-80.08***
	[-122.0,-38.11]
Decreasing	-9.832
	[-48.04,28.38]
Random	-52.52***
	[-80.04,-24.99]
Group Contribution	0.00240
	[-0.319,0.324]
Increasing × Group Contribution	0.446
	[-0.259,1.151]
Decreasing × Group Contribution	-0.201
	[-0.777,0.376]
Random × Group Contribution	0.0812
	[-0.265,0.427]
Constant	174.0***
	[156.9,191.2]
Observations	1920
Clusters	24

Notes: Individual stage contribution on treatment and group stage contributions. Last 10 rounds. Wild cluster bootstrapping 95% confidence intervals in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

slightly in group stage contributions for every treatment except **Decreasing**, but the effects are significantly different from null only for **Increasing**.[22] There are also no significant differences in the effects between treatments.[23]. Combining these results with the robustness tests above suggests that mere differences in group stage contributions don't drive the differences in individual stage contributions.

5.2.3 Within-Group Coordination and Asymmetry

I now more closely inspect the nature of the individual stage asymmetry that arises in each treatment. Understanding how asymmetry differs across treatments may shed light on differences in individual stage contributions. Recall that asymmetry in the individual stage arises when group mates select different group stage contributions or are randomly assigned different strengths. I first analyze the extent to which group members coordinate contributions in the group stage.

Figure 8 shows a scatter plot in which each dot represents one group of two contestants and captures the ratio of the minimum group stage contribution to the maximum group stage

[22] Wald tests comparing constants in linear regression to predictions: INC, $p = 0.09$; DEC, $p = 0.51$; SYM, $p = 0.99$; $RAND$, $p = 0.30$

[23] Wald tests comparing constants in linear regression: INC vs. DEC, $p = 0.18$; INC vs. SYM, $p = 0.28$; INC vs. $RAND$, $p = 0.37$; DEC vs. SYM, $p = 0.51$; DEC vs. $RAND$, $p = 0.29$; SYM vs $RAND$, $p = 0.68$

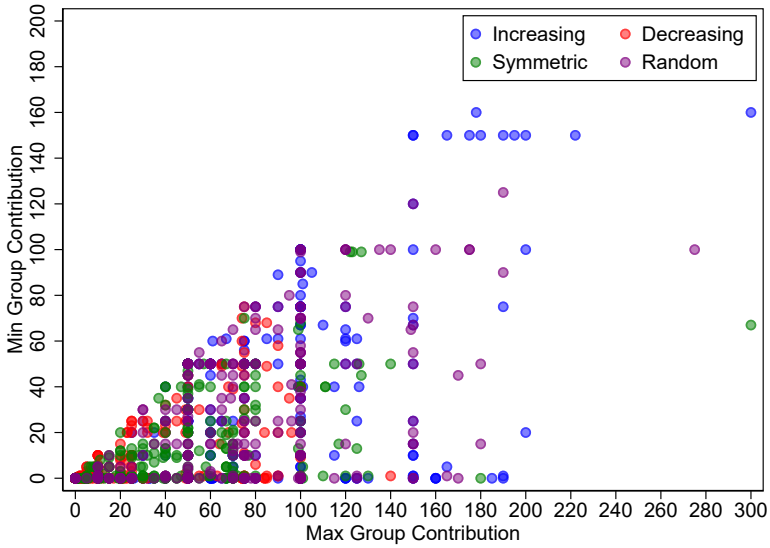


Figure 8: Group Coordination, Last 10 Rounds

contribution within a given group. Points along the diagonal represent perfect coordination in which both group members choose the same contribution level. Points along the horizontal axis represent perfect free-riding in which one group member contributes something positive while the other contributes 0. Points in between represent partial free-riding in which both group members contribute something positive but one out-contributes the other. In the treatments with endogenous strength, the theory predicts perfect coordination. In **Symmetric**, many equilibria exist involving asymmetric contributions. The scatter plot reveals that, while some perfect coordination and perfect free-riding exists, most groups involve group members choosing different positive group stage contributions. The same pattern arises regardless of treatment.

In the endogenous strength treatments, asymmetric group stage contributions translate to asymmetric strengths in the individual stage. In **Random**, asymmetry is determined by a random draw. Figure 9 shows how differences in group stage contributions flow to individual stage asymmetry. I plot the mean within-group strength ratio for the individual stage in the last 10 rounds. Recall that strength ratio is the ratio of the group's minimum strength to the group's maximum strength. Mechanically, in **Symmetric**, since both group members are assigned a strength of 1 in the individual stage, strength ratio equals 1.

Theory predicts equal strengths in the endogenous strength treatments. I instead observe that in both endogenous strength treatments, the weaker group member is about 2/3 as strong as the stronger group member upon entering the individual stage. This corresponds with the average draw of strengths in **Random**. The strength ratio is slightly higher on average in **Decreasing** compared to **Increasing** and **Random**, but the difference is only significant when compared to **Random**.[24] In appendix B, I plot histograms of strength ratios by treatment which reveal slightly more mass at lower strength ratios in **Increasing** even though there is no difference on average.

Result 6 (Asymmetry in the individual stage)**.** *Contrary to theory, subjects in the same group*

[24]Wilcoxon rank-sum test: INC vs. DEC, $p = 0.18$; INC vs. $RAND$, $p = 0.31$; DEC vs. $RAND$, $p = 0.03$;

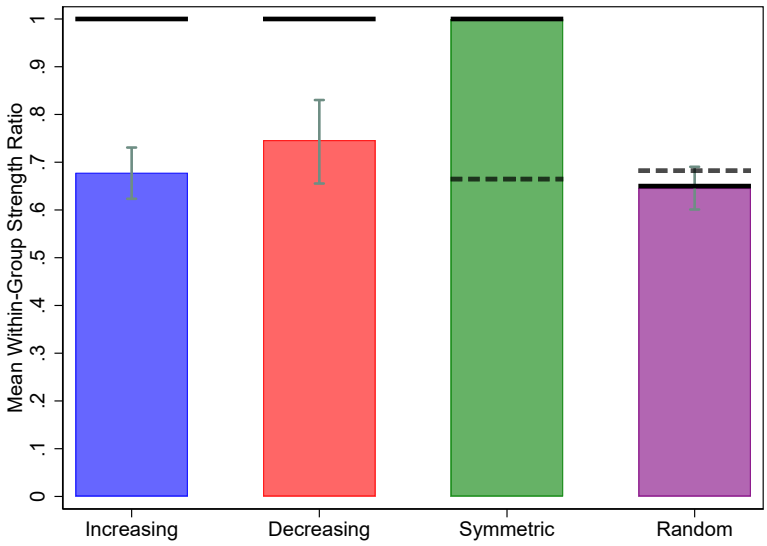


Figure 9: Strength Ratio, Last 10 Rounds

submit different group stage contributions. This induces off equilibrium path asymmetry in the individual stage of the endogenous strength treatments. On average, the strength ratio is not different between the endogenous strength treatments.

In figure 9, I also plot a dashed line for each exogenous strength treatment representing the strength ratio that would've resulted if strength were endogenous. This again illustrates the similarity of the distribution of group stage contributions within groups across all four treatments.

5.2.4 Asymmetric Contests

In this final section, I delve deeper into the effects of asymmetry on individual stage contributions and explore whether the effects differ across treatments that vary how asymmetry arises. Once subjects arrive in the individual stage and face an asymmetric contest, it should not matter how they got there. In theory, a subject who is twice as strong as her group mate shouldn't behave differently whether she is strong due to high group stage effort, low group stage effort, or a random draw. Any differences in effort should be explained only by differences in the degree of asymmetry between contestants within a group. Unexplained variation must be due to some behavioral biases regarding how asymmetry arises.[25]

I first study the effect of relative strength on contributions across the three treatments where asymmetry exists. I define relative strength as own strength divided by group mate strength and calculate it for each subject in each individual stage contest. Table 5 regresses individual

[25]Strictly speaking, in the endogenous strength treatments, there is intentionally and mechanically not random variation in strengths or measures of asymmetry. Subjects in these treatments certainly select into their strengths. However, once the individual stage arises, group stage behavior is sunk, and the realized strengths should in theory be treated no differently than if they were randomly assigned. Any differences that arise suggest some selection effects or other behavioral effects of endogenously assigning strengths. These biases are exactly what the following regression analyses seek to explore.

Table 5: Individual Stage Contributions on Relative Strength, Last 10 Rounds

	(1)
Increasing	-1.902
	[-33.19,29.38]
Decreasing	33.69
	[-40.25,107.6]
Relative Strength	11.01*
	[1.197,20.82]
Increasing × Relative Strength	-2.737
	[-19.27,13.80]
Decreasing × Relative Strength	1.869
	[-69.38,73.12]
Constant	112.8***
	[91.81,133.7]
Observations	1440
Clusters	18

Notes: Regression of individual stage contributions on relative strength and treatment dummies. Last 10 rounds. Wild cluster bootstrapping 95% confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

stage contributions on relative strength and a treatment dummy. I find relative strength only significantly impacts individual stage contributions in **Random** where a higher relative strength increases contributions.[26] I find no significant difference between treatments.[27]

This suggests that all players respond to relative strength in the same way and that relative strength only significantly impacts contributions when strengths are exogenously assigned. However, the relative strength measure may conflate two distinct elements of asymmetry: 1) the degree of asymmetry in the contest and 2) the discrete coding of whether a player is stronger or weaker. Theory predicts contributions should only be affected by the degree of asymmetry and not by whether a player is weaker or stronger. As the degree of asymmetry increases, contributions should decrease equally for both players regardless of position. Studying relative strength may conflate the effects of asymmetry and position on contributions. I next try to look at each of these effects separately.[28]

First, I ignore the degree of asymmetry and study the effect of being stronger or weaker relative to group mate. I can compare this analysis to previous research on asymmetric contests (particularly Fonseca (2009)) and explore whether results differ when asymmetry arises endogenously.

[26] Wald tests comparing constants in linear regressions to null: INC, $p = 0.26$; DEC, $p = 0.72$; $RAND$, $p = 0.06$

[27] Wald tests comparing constants in linear regression: INC vs. DEC, $p = 0.91$; INC vs. $RAND$, $p = 0.76$; DEC vs. $RAND$, $p = 0.97$

[28] In tables 24 and 25, I offer an alternative approach to try to separate the effects of asymmetry and position in one regression. I regress individual stage contributions on strength ratio and a dummy for whether the player is stronger or weaker and report the slope coefficients on strength ratio for each treatment-type combination. In doing so, I allow the effects of asymmetry to differ between stronger and weaker players and across treatments.

Table 6: Individual Stage Contributions by Treatment and Type, Last 10 Rounds

Type	Increasing	Decreasing	Symmetric	Random
Equal	146.74	133.60	174.10	126.14
	[71.29,178.82]	[96.15,182.28]	[151.58,188.34]	[84.37,159.25]
Strong	133.20	181.69	.	139.19
	[105.95,168.52]	[118.02,236.77]		[115.63,165.16]
Weak	99.78	154.78	.	112.46
	[73.93,131.66]	[111.80,206.29]		[87.90,140.90]

Notes: Estimates from regressing individual stage contribution on treatment dummy and type dummy. Last 10 rounds. 95% wild cluster bootstrap confidence intervals in brackets.

Fonseca (2009) compares symmetric and asymmetric contests where strength is randomly assigned for each player. Each treatment involves either a symmetric or an asymmetric format, and subjects within treatments are categorized as being equal, weak, or strong. The author draws three main conclusions from the experimental data. Firstly, subjects in symmetric contests and both types in asymmetric contests all overinvest on average. Secondly, both types in asymmetric contests invest less on average than subjects in symmetric contests. Thirdly, both types in asymmetric contests invest about the same level on average.

I compare individual stage contributions and percent deviations from stage Nash equilibrium across treatment and type (equal, strong, or weak). Table 6 shows mean contributions by treatment and type and table 7 shows mean percent deviation. In these tables, equality is defined broadly as being within a range based on the absolute strength values in each treatment (within 0.2 in **Increasing** and **Random** and within 0.02 in **Decreasing**). In tables 20 and 21 in Appendix B, I consider only exact equality, and the qualitative results do not change.

To test whether overinvestment occurs, I test where percent deviation from Nash equilibrium differs from 0. Contrary to previous research, I find that not all types exhibit overinvestment. Equal types in both endogenous strength treatments and the weak types from **Increasing** and **Random** actually underinvest on average. Strong types and subjects in **Symmetric** (equal by default) don't deviate from Nash equilibrium on average or slightly overinvest. It appears the lack of overinvestment we observe is driven by the weak and equal types. Deviations are statistically significant for **Symmetric** ($p = 0.04$) and the weak types in **Increasing** ($p = 0.07$) and **Random** ($p = 0.08$).

Result 7 (Underinvestment by weak and equal types)**.** *The surprising lack of overinvestment I observe is driven by weak and equal types in asymmetric contests.*

Fonseca (2009) secondly finds that both types in asymmetric contests invest less than those in symmetric contests. In my experiment, I have symmetric contests that arise exogenously and endogenously, so I compare behavior in asymmetric contests to each in turn. Contributions by both strong and weak types in **Increasing** and **Random** are significantly lower than contributions in **Symmetric**, while neither type in **Decreasing** contributes differently.[29] Contributions in an endogenously symmetric contest are significantly lower in both endogenous strength treat-

[29]Wald tests comparing constants in linear regression: INC_{strong} , $p = 0.03$; INC_{weak} , $p = 0.004$; DEC_{strong} , $p = 0.74$; DEC_{weak} , $p = 0.33$; $RAND_{strong}$, $p = 0.02$; $RAND_{weak}$, $p = 0.002$

Table 7: Individual Stage Percent Deviation by Treatment and Type, Last 10 Rounds

Type	Increasing	Decreasing	Symmetric	Random
Equal	-0.02	-0.11	0.16	-0.16
	[-0.52,0.19]	[-0.36,0.22]	[0.01,0.26]	[-0.44,0.06]
Strong	0.01	0.29	.	0.03
	[-0.22,0.25]	[-0.18,0.72]		[-0.14,0.23]
Weak	-0.28	0.09	.	-0.18
	[-0.50,0.02]	[-0.22,0.42]		[-0.37,0.02]

Notes: Estimates from regressing percent deviation from individual stage NE on treatment dummy and type dummy. Last 10 rounds. 95% wild cluster bootstrap confidence intervals in brackets.

ments on average compared to contributions in an exogenously symmetric contest. [30] Therefore, the results differ when comparing contributions in endogenously asymmetric contests to endogenously symmetric contests as a benchmark. In **Increasing**, there is no longer any significant difference for either type, while in **Decreasing**, contributions by strong and weak types are actually *higher* and significantly so for the strong types.[31]

Regardless of which symmetric contest benchmark I use (exogenously symmetric or endogenously symmetric), contrary to theoretical predictions and previous experimental findings, I do not find that both strong and weak types in asymmetric contests always contribute less on average than those in symmetric contests. In particular, both types in **Decreasing** compete similarly to if they were symmetric.

Result 8 (Asymmetric versus symmetric contests)**.** *Contrary to theoretical predictions and previous research, asymmetry does not always decrease competition from the level found in symmetric contests.*

Comparing types within treatments, the strong player contributes slightly more on average than the weak player in all three treatments involving asymmetry. The difference is statistically significant in **Random**.[32] Equality across types is the prediction of theory and the empirical observation of Fonseca (2009).

Result 9 (Strong vs. weak)**.** *Contrary to theory and previous research, I find suggestive evidence that stronger players contribute more on average than weaker players in asymmetric contests.*

I next compare within types across treatments. Previous research on asymmetric contests has been unable to perform this analysis because it holds fixed the method for assigning types. I observe that within strong types, there are no differences between **Increasing** and **Random**. However, strong types contribute more in **Decreasing** on average compared to both **Increasing** and **Random**.[33] Similarly, for weak types, there are no differences between **Increasing**

[30] Wald tests comparing constants in linear regression: INC_{equal} , $p = 0.03$; DEC_{equal} , $p = 0.03$

[31] Wald tests comparing constants in linear regression: INC_{strong} , $p = 0.98$; INC_{weak} , $p = 0.34$; DEC_{strong} , $p = 0.08$; DEC_{weak} , $p = 0.24$

[32] Wald tests comparing constants in linear regression: INC_{strong} vs. INC_{weak}, $p = 0.21$; DEC_{strong} vs. DEC_{weak}, $p = 0.28$; $RAND_{strong}$ vs. $RAND_{weak}$, $p = 0.01$

[33] Wald tests comparing constants in linear regression: INC_{strong} vs. DEC_{strong}, $p = 0.12$; INC_{strong} vs. $RAND_{strong}$, $p = 0.79$; DEC_{strong} vs. $RAND_{strong}$, $p = 0.14$

and **Random**, but weak types contribute more in **Decreasing** on average compared to both **Increasing** and **Random**, and these differences are statistically significant.[34]

Across the board, strong and weak types in **Decreasing** contribute more on average than their counterparts in other treatments. When higher strength is earned through shirking as opposed to hard work or random assignment, it appears overall competitiveness remains high. On the other hand, competitiveness dampens when a high position is earned through hard work or given exogenously.

My results resemble a nuanced expression of the entitlement effect first observed in dictator and ultimatum games (Hoffman et al., 1994; Cappelen et al., 2007; Oxoby & Spraggon, 2008). Typically, individuals behave less generously when they earn an advantaged position compared to when it is given. In my competitive setting, I find that people also seem to care about *how* an advantaged position is earned. When the strong position is earned through high group contribution, both players appear to accept the outcome and reduce competition. This is especially true for weak players. Players also seem to accept the strength distribution as fair and dampen competition when strengths are randomly assigned. However, when the strong position is earned through a lack of group contribution, then competition ramps up. It appears neither player accepts that the stronger group member is entitled to the prize. Contrary to previous research, individuals don't care as much about the distinction between earned versus given advantage, but they do respond to differences in how advantage is earned.

Interestingly, equal types in **Decreasing** do not contribute significantly more on average than their counterparts in **Increasing**, and both equal types in endogenous strength treatments contribute significantly less than subjects in **Symmetric**[35]. When individuals are evenly matched due to endogenous coordination, they behave less competitively against each other on average compared to when they are evenly matched exogenously.

Result 10 (Comparing types across treatments)**.** *Both stronger and weaker types compete more fiercely when high strength is earned through shirking rather than hard work or random assignment. Competition in symmetric contests is dampened when symmetry arises through prior coordination compared to when it is exogenously assigned.*

Results 7, 8, and 9 show how the behavior in the asymmetric contests in my experiment sometimes differs markedly from the findings of previous research, and result 10 provides new insights into competition in asymmetric contests. Contrary to theoretical predictions, I find differences in behavior across treatments. Altogether these results provide some evidence that people in asymmetric situations are indeed sensitive to the origin of the asymmetry.

Next, I ignore types and focus only on the effect of the degree of asymmetry. Recall that theory predicts that the equilibrium level of contributions decreases as a contest becomes more asymmetric. To study this and compare across treatments, I regress individual stage contributions on treatment dummies and strength ratio in table 8. Both members of a group have the same measure for strength ratio, so this regression truly only captures the effect of the degree of asymmetry and doesn't take into account whether a player is stronger or weaker.

[34] Wald tests comparing constants in linear regression: INC_{weak} vs. DEC_{weak}, $p = 0.03$; INC_{weak} vs. $RAND_{weak}$, $p = 0.65$; DEC_{weak} vs. $RAND_{weak}$, $p = 0.04$

[35] Wald tests comparing constants in linear regression: INC_{equal} vs. DEC_{equal}, $p = 0.72$; INC_{equal} vs. SYM, $p = 0.03$; DEC_{equal} vs. SYM, $p = 0.03$

Table 8: Individual Stage Contributions and Strength Ratio, Last 10 Rounds

	(1)
Increasing	-54.27
	[-137.7,29.20]
Decreasing	74.11
	[-36.50,184.7]
Strength Ratio	58.78
	[-23.30,140.9]
Increasing × Strength Ratio	70.10
	[-43.95,184.2]
Decreasing × Strength Ratio	-60.46
	[-197.6,76.66]
Constant	87.87**
	[32.83,142.9]
Observations	1440
Clusters	18

Notes: Regression of individual stage contributions on strength ratio and treatment dummies. Last 10 rounds. Wild cluster bootstrapping 95% confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

I compare coefficients on strength ratio to 0,[36] and I compare across treatments.[37] For both **Increasing** and **Random**, a higher strength ratio (greater symmetry) is positively correlated with individual stage contributions. This matches the theoretical prediction that greater symmetry induces more fierce competition. Contrary to theory, in **Decreasing** individual stage contributions decrease in strength ratio. The difference in the effect of strength ratio is significant when comparing the two endogenous strength treatments.

These results provide further evidence that individuals in asymmetric situations react differently depending on how asymmetry arises. In both endogenous strength treatments in my experiment, a gap in strengths arises due to differences in contributions to group effort. When contributing more to the group earns high strength, the greater the asymmetry, the less aggressive the competition. Again, it seems competitors in this case accept that the higher group contributor is entitled in some way to the higher likelihood of winning the prize. I see similar results when strengths are randomly assigned. In either case, players are more inclined to accept the strength distribution and compete less fiercely as a result. On the other hand, when the shirker in the group attains higher strength, players behave more competitively the wider the strength gap becomes. In this case, players don't accept that the stronger player has earned the right to the prize and still compete aggressively. The more severe the asymmetry, the more aggressive the competition. Once again, the main contextual factor which affects behavior is how the advantaged position is earned.

[36] Wald tests comparing constants in linear regression to predictions: INC, $p = 0.03$; DEC, $p = 0.98$; $RAND$, $p = 0.21$

[37] Wald tests comparing constants in linear regression to predictions: INC vs. DEC, $p = 0.09$; INC vs. $RAND$, $p = 0.26$; DEC vs. $RAND$, $p = 0.41$

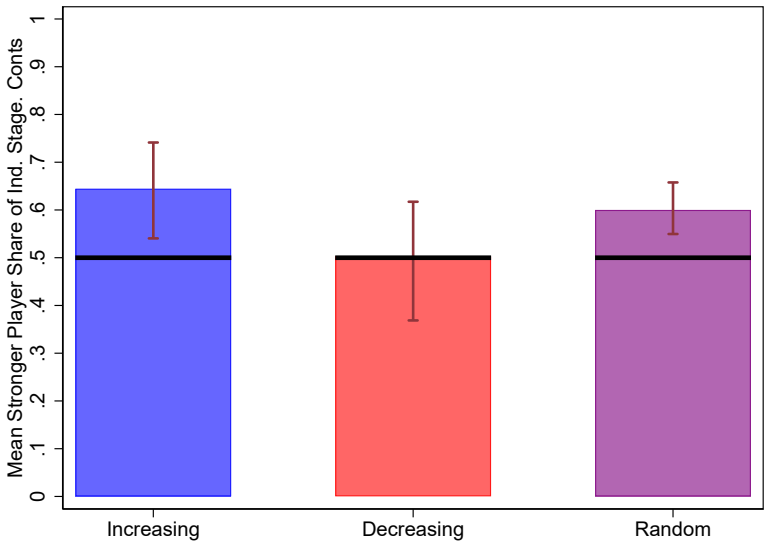


Figure 10: Share of Individual Stage Contributions, Last 10 Rounds

Result 11 (Effect of the degree of asymmetry on competition)**.** *Greater asymmetry affects competition differently depending on how asymmetry arises. When a strong position is earned through high effort or randomly assigned, greater asymmetry dampens competition. On the contrary, when a strong position is earned through shirking, greater asymmetry heightens competition.*

Finally, I compare across treatments the share of individual stage contributions made by the strong type. In figure 10, I plot the stronger player's share of the total individual stage contributions in only asymmetric contests. Theory predicts equal individual stage contributions for both contestants even if asymmetry arises and no differences between treatments.

In **Decreasing**, the stronger and weaker player both contribute about an equal share of the total individual stage contributions. However, in **Increasing** and **Random**, the stronger player behaves relatively more aggressively, contributing about 64% and 60%, respectively, of the total individual stage contributions on average. In both treatments, the stronger player contributes a share which is significantly greater than an even split.[38] The stronger player's share is significantly higher in **Increasing** compared to **Decreasing**.[39]

In both endogenous strength treatments, asymmetry arises from a lack of coordination in the group stage. The degree of asymmetry is equivalent on average across both treatments. However, the source of the asymmetry from an individual's perspective varies depending on the direction of the strength function. In **Increasing**, a stronger subject earns her position from contributing more in the group stage. On the contrary, a stronger subject in **Decreasing** earns her position by contributing less in the group stage. The difference in how asymmetry arises affects how stronger versus weaker players compete. When a subject earns an advantage by outworking her group mate, she competes relatively aggressively in the individual stage. When a subject receives an advantage after being outworked, she matches her weaker group mate's

[38] Wald tests comparing constants in linear regression: INC, $p = 0.04$; DEC, $p = 0.96$; $RAND$, $p = 0.01$

[39] Wilcoxon rank-sum test: INC vs. DEC, $p = 0.04$; INC vs. $RAND$, $p = 0.24$; DEC vs. $RAND$, $p = 0.18$

Table 9: Strong Player Share of Individual Stage Contributions on Strength Ratio, Last 10 Rounds

	(1)
Increasing	0.327*
	[0.0774,0.577]
Decreasing	-0.259
	[-0.849,0.332]
Strength Ratio	-0.145
	[-0.291,0.00176]
Increasing × Strength Ratio	-0.436**
	[-0.690,-0.182]
Decreasing × Strength Ratio	0.242
	[-0.475,0.959]
Constant	0.694***
	[0.577,0.810]
Observations	657
Clusters	18

Notes: Regression of stronger player share of individual stage contributions on strength ratio and treatment dummies. Last 10 rounds. Wild cluster bootstrapping 95% confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

competitiveness in the individual stage. This provides more evidence for the implication of result 11 which suggests that subjects accept the stronger player's claim to the prize when the strong position is earned through high group contribution or randomly assigned but not when the strong position is earned through shirking.

In table 9, I also regress the stronger player's share of individual stage contributions on treatment and strength ratio. I compare the slope coefficients on strength ratio to 0,[40] and I compare across treatments.[41]

I find a significant difference in the effect of strength ratio on the stronger player's share of contributions between the two endogenous strength treatments and between **Increasing** and **Random**. When a subject earns a strong position in **Increasing**, greater asymmetry increases her share of the contributions. The effect is stronger than when a subject becomes strong exogenously as in **Random**. On the contrary, stronger players who receive a strong position by being outworked as in **Decreasing** actually submit a lower share of contributions as the strength gap widens.

Result 12 (Stronger player's share of individual stage contributions)**.** *On average, the share of individual stage contributions expended by the stronger player is greater when strength is increasing versus when it is decreasing. As contests become more asymmetric, the stronger player's share increases most strongly when strength is earned through high contributions, increases less*

[40] Wald tests comparing constants in linear regression: INC, $p = 0.009$; DEC, $p = 0.84$; $RAND$, $p = 0.09$

[41] Wald tests comparing constants in linear regression: INC vs. DEC, $p = 0.09$; INC vs. $RAND$, $p = 0.006$; DEC vs. $RAND$, $p = 0.63$

strongly when strength is randomly assigned, and decreases when strength is earned through low contributions.

The results on the asymmetric contests in this experiment suggests that individuals respond differently to asymmetry depending on how that asymmetry arises. I observe a nuanced entitlement effect in this environment which drives people to compete differently depending on whether asymmetry arises endogenously or exogenously and especially whether an advantaged position is earned through high effort or shirking.

6 Conclusion

In this paper, I study situations in which individuals coordinate and compete for prizes. Such situations commonly arise in the real world in organizations, sports, markets and political economy. I focus on the link between an individual's contribution to a group's collective effort and her position in competition within the group. To study this topic, I develop a two-stage model of intergroup and intragroup competition. In the group stage, two groups of two agents each compete in a group Tullock contest for the right to advance to the next stage. Group members coordinate and pool their contributions to increase their likelihood of advancing. The group that wins moves on to the individual stage in which group members compete against one another for an indivisible prize. Contributions in the individual stage are multiplied by strength. Strength may be exogenously or endogenously determined and symmetric or asymmetric. Endogenous strength depends on an individual's contributions in the group stage. In one case, higher group stage contributions translate into higher strength, while in another case higher group stage contributions translate into lower strength. I solve the model for different methods of strength assignment and develop testable hypotheses regarding equilibrium contributions in each stage and comparative statics.

I then conduct a laboratory experiment in which subjects play the two-stage game for 20 repeated rounds. I analyze the data on contributions in each stage and find three main sets of results. First, I find little evidence of overinvestment in the individual stage. This goes directly against robust historical findings and potentially encourages testing the predictions of contest theory in more complex and realistic dynamic environments. Secondly, within each stage, contributions differ depending on the method for assigning strengths, while total contributions over both stages remain relatively constant. Consequently, the optimal method for assigning individual strengths from society or a principal's perspective depends on whether collective or individual effort is considered productive versus wasteful.

Finally, in analyzing the individual stage in my experiment, I am able to study asymmetric contests with different origins of asymmetry. Contrary to theoretical predictions, I find that individuals competing in asymmetric Tullock contests are indeed sensitive to how asymmetry arises. In particular, the distribution of strengths isn't the only factor that affects contribution decisions. Individuals also care about whether strength depends on group contributions and how exactly it is manipulated. Specifically, I find that individuals behave more competitively when the low group contributors possess an advantage compared to when advantage is randomly assigned or earned through high group contributions. This finding adds nuance to the commonly observed entitlement effect in previous research on other-regarding preferences where individuals mainly care about whether property rights are given or earned. Within firms and other organizations, unevenly matched agents frequently interact and often compete. According

to my results, managers and policymakers overseeing such situations should pay close attention to the origins of asymmetry and adapt their strategies accordingly to optimize the effectiveness of their agents.

This paper is the first to theoretically and experimentally study the relationship between group contributions and individual strength in competition between and within groups. The results advance the vast literature on group and individual contests. Even more importantly, this research offers new insights into the behavioral biases individuals possess towards the assignment of property rights or other advantages.

Altogether, this paper provides new insights into how groups of individuals coordinate and compete within organizations. Future research should explore different asymmetric settings to further test how people react when asymmetry arises for different reasons. Other follow up work could seek to identify optimal policy responses to the behavioral biases I recognize.

References

Abbink, K., Brandts, J., Herrmann, B., & Orzen, H. (2010). Intergroup conflict and intra-group punishment in an experimental contest game. *American Economic Review*, *100*(1), 420–447.

Ahn, T.-K., Isaac, R. M., & Salmon, T. C. (2011). Rent seeking in groups. *International Journal of Industrial Organization*, *29*(1), 116–125.

Altmann, S., Falk, A., & Wibral, M. (2012). Promotions and incentives: The case of multistage elimination tournaments. *Journal of Labor Economics*, *30*(1), 149–174.

Baik, K. H. (1994). Effort levels in contests with two asymmetric players. *Southern Economic Journal*, 367–378.

Bhattacharya, P. (2016). Inter-team contests with power differential. *Journal of Economic Behavior & Organization*, *132*, 157–175.

Bhattacharya, P., & Rampal, J. (2024). Contests within and between groups: Theory and experiment. *Games and Economic Behavior*, *145*, 467–492.

Brookins, P., Lightle, J. P., & Ryvkin, D. (2015). An experimental study of sorting in group contests. *Labour Economics*, *35*, 16–25.

Cappelen, A. W., Hole, A. D., Sørensen, E. Ø., & Tungodden, B. (2007). The pluralism of fairness ideals: An experimental approach. *American Economic Review*, *97*(3), 818–827.

Cason, T. N., Sheremeta, R., & Zhang, J. (2017). Asymmetric and endogenous communication in competition between groups. *Experimental Economics*.

Cason, T. N., Sheremeta, R. M., & Zhang, J. (2012). Communication and efficiency in competitive coordination games. *Games and economic behavior*, *76*(1), 26–43.

Cherry, T. L., Frykblom, P., & Shogren, J. F. (2002). Hardnose the dictator. *American economic review*, *92*(4), 1218–1221.

Choi, J. P., Chowdhury, S. M., & Kim, J. (2016). Group contests with internal conflict and power asymmetry. *The Scandinavian Journal of Economics*, *118*(4), 816–840.

Clark, D. J., & Riis, C. (1998). Competition over more than one prize. *The American Economic Review*, *88*(1), 276–289.

Dechenaux, E., Kovenock, D., & Sheremeta, R. M. (2015). A survey of experimental research on contests, all-pay auctions and tournaments. *Experimental Economics*, *18*(4), 609–669.

Fallucchi, F., Fatas, E., Kölle, F., & Weisel, O. (2021). Not all group members are created equal: Heterogeneous abilities in inter-group contests. *Experimental Economics*, *24*(2), 669–697.

Fischbacher, U. (2007). z-tree: Zurich toolbox for ready-made economic experiments. *Experimental economics*, *10*(2), 171–178.

Fonseca, M. A. (2009). An experimental investigation of asymmetric contests. *International Journal of Industrial Organization*, *27*(5), 582–591.

Franke, J., Kanzow, C., Leininger, W., & Schwartz, A. (2013). Effort maximization in asymmetric contest games with heterogeneous contestants. *Economic Theory*, *52*(2), 589–630.

Fu, Q., & Lu, J. (2012). The optimal multi-stage contest. *Economic Theory*, *51*(2), 351–382.

Gradstein, M., & Konrad, K. A. (1999). Orchestrating rent seeking contests. *The Economic Journal*, *109*(458), 536–545.

Greiner, B. (2015). Subject pool recruitment procedures: organizing experiments with orsee. *Journal of the Economic Science Association*, *1*(1), 114–125.

Hoffman, E., McCabe, K., Shachat, K., & Smith, V. (1994). Preferences, property rights, and anonymity in bargaining games. *Games and Economic behavior*, *7*(3), 346–380.

Katz, E., & Tokatlidu, J. (1996). Group competition for rents. *European Journal of Political Economy*, *12*(4), 599–607.

Kirkegaard, R. (2012). Favoritism in asymmetric contests: Head starts and handicaps. *Games and Economic Behavior*, *76*(1), 226–248.

Lazear, E. P., & Rosen, S. (1981). Rank-order tournaments as optimum labor contracts. *Journal of political Economy*, *89*(5), 841–864.

Münster, J. (2007). Simultaneous inter-and intra-group conflicts. *Economic Theory*, *32*(2), 333–352.

Münster, J. (2009). Group contest success functions. *Economic Theory*, *41*(2), 345–357.

Oxoby, R. J., & Spraggon, J. (2008). Mine and yours: Property rights in dictator games. *Journal of Economic Behavior & Organization*, *65*(3-4), 703–713.

Rogerson, W. (1982). The social costs of monopoly and regulation: A game-theoretic analysis. *Bell Journal of Economics*, *13*(2), 391–401.

Ryvkin, D. (2011). Fatigue in dynamic tournaments. *Journal of Economics & Management Strategy*, *20*(4), 1011–1041.

Schmitt, P., Shupp, R., Swope, K., & Cadigan, J. (2004). Multi-period rent-seeking contests with carryover: Theory and experimental evidence. *Economics of Governance*, *5*(3), 187–211.

Schotter, A., & Weigelt, K. (1992). Asymmetric tournaments, equal opportunity laws, and affirmative action: Some experimental results. *The Quarterly Journal of Economics*, *107*(2), 511–539.

Sheremeta, R. M. (2010a). Expenditures and information disclosure in two-stage political contests. *Journal of Conflict Resolution*, *54*(5), 771–798.

Sheremeta, R. M. (2010b). Experimental comparison of multi-stage and one-stage contests. *Games and Economic Behavior*, *68*(2), 731–747.

Sheremeta, R. M. (2011). Perfect-substitutes, best-shot, and weakest-link contests between groups. *Korean Economic Review*, *27*, 5–32.

Sheremeta, R. M. (2018). Behavior in group contests: A review of experimental research. *Journal of Economic Surveys*, *32*(3), 683–704.

Skaperdas, S. (1996). Contest success functions. *Economic theory*, *7*(2), 283–290.

Stein, W. E. (2002). Asymmetric rent-seeking with more than two contestants. *Public Choice*, *113*(3), 325–336.

Stein, W. E., & Rapoport, A. (2004). Asymmetric two-stage group rent-seeking: Comparison of two contest structures. *Public Choice*, *118*(1), 151–167.

Tsoulouhas, T., Knoeber, C. R., & Agrawal, A. (2007). Contests to become ceo: incentives, selection and handicaps. *Economic Theory*, *30*(2), 195–221.

Tullock, G. (1967). The welfare costs of tariffs, monopolies, and theft. *Economic Inquiry*, *5*(3), 224–232.

Tullock, G. (1980). Efficient rent seeking. In *Toward a theory of the rent-seeking society* (p. 97-112). Texas A&M University Press.

Wärneryd, K. (1998). Distributional conflict and jurisdictional organization. *Journal of public economics*, *69*(3), 435–450.

A Additional Theoretical Results and Proofs

A.1 Individual stage equilibrium with binding budget constraints

This appendix characterizes the Nash equilibrium of the individual stage when players may be budget constrained. Suppress group indices for this analysis. Let remaining endowments after the group stage be

$$r_i \equiv w - x_i \geq 0, \qquad i \in \{1, 2\},$$

and let strengths be (θ_1, θ_2), where $\theta_i > 0$. In the individual stage, agents choose $y_i \in [0, r_i]$ to maximize

$$u_i(y_i, y_j) = \frac{\theta_i y_i}{\theta_i y_i + \theta_j y_j} V - y_i + r_i, \tag{20}$$

The unconstrained first-order condition for agent i given the contribution of agent j is

$$\frac{\partial u_i}{\partial y_i} = \frac{V \theta_i \theta_j y_j}{(\theta_i y_i + \theta_j y_j)^2} - 1 = 0,$$

and the unconstrained best response function is

$$y_i(y_j) = \frac{\sqrt{V \theta_i \theta_j y_j} - \theta_j y_j}{\theta_i}. \tag{21}$$

Imposing the budget constraint $y_i \leq r_i$ implies the constrained best response

$$BR_i(y_j) = \min \left\{ r_i,\ \max \left\{ y_i(y_j) \right\} \right\}. \tag{22}$$

If budgets do not bind for either player, the unique interior Nash equilibrium is the standard outcome characterized in Lemma 1:

$$y_1^* = y_2^* = \frac{V \theta_1 \theta_2}{(\theta_1 + \theta_2)^2}. \tag{23}$$

Suppose the budget constraint binds only for agent 1. Then the Nash equilibrium satisfies

$$y_1^* = r_1, \qquad y_2^* = \frac{\sqrt{V \theta_1 \theta_2 r_1} - \theta_1 r_1}{\theta_2}, \tag{24}$$

Suppose both agents are constrained. Then both constraints bind in equilibrium:

$$y_1^* = r_1, \qquad y_2^* = r_2. \tag{25}$$

Note that in our experiment, budget constraints never bind on the equilibrium path. However, budget constraints may bind off the equilibrium path. Deviating to a group stage contribution that makes budget constraints bind in the individual stage is unprofitable. Therefore, even if budget constraints can bind in the individual stage for certain group stage contributions, the main analysis of the subgame-perfect equilibria given my experimental parameters remains unchanged.

B Supplementary Results

Table 10: Summary Statistics, All Rounds

	Group Stage		Individual Stage	
Treatment	Mean	SPE	Mean	SPE
Increasing	68.41	42.09	125.05	150.00
	[42.02, 98.27]		[108.97, 151.44]	
Decreasing	25.37	4.19	147.18	150.00
	[11.64, 45.94]		[115.95, 174.20]	
Symmetric	42.63	18.75	162.79	150.00
	[34.42, 49.87]		[138.62, 175.00]	
Random	59.66	27.55	123.22	143.25
	[39.64, 82.27]		[99.50, 143.68]	

Notes: Mean contributions by stage and treatment. All rounds. 95% wild cluster bootstrap confidence intervals in brackets.

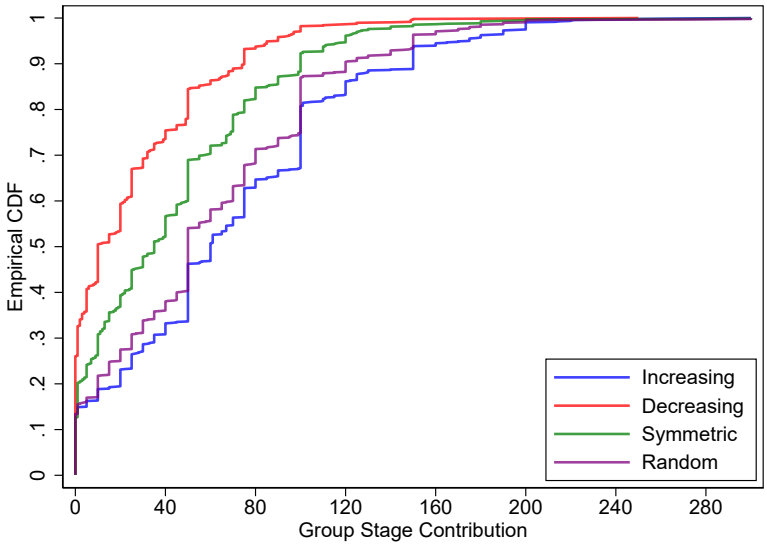


Figure 11: Distribution of Group Stage Contributions, All Rounds

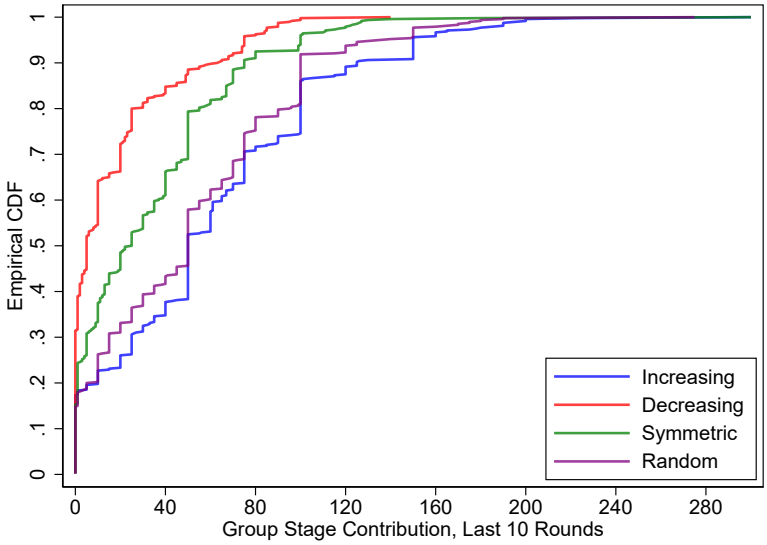


Figure 12: Distribution of Group Stage Contributions, Last 10 Rounds

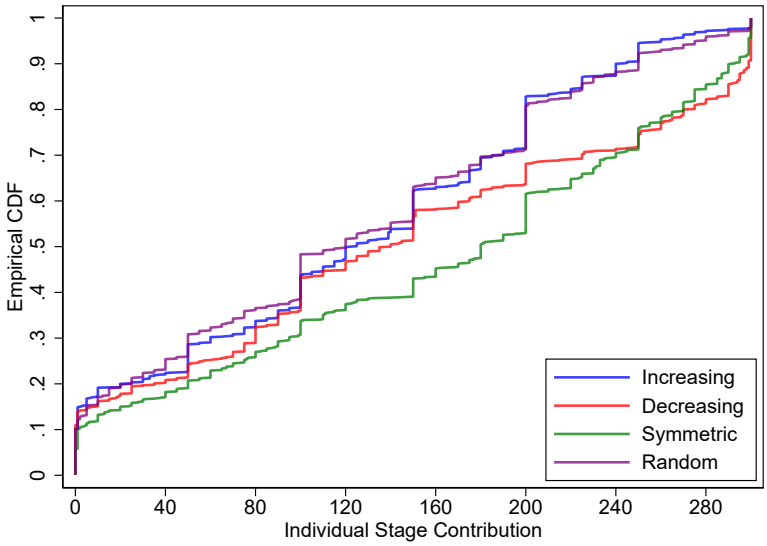


Figure 13: Distribution of Individual Stage Contributions, All Rounds

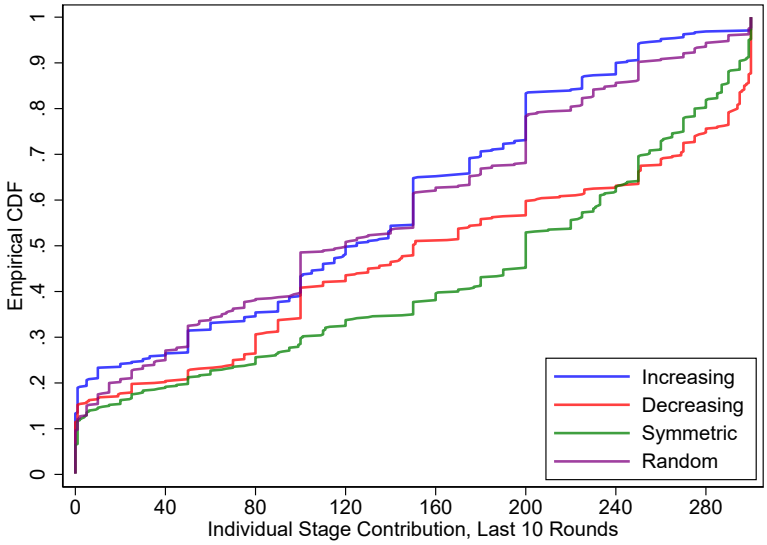


Figure 14: Distribution of Individual Stage Contributions, Last 10 Rounds

Table 11: Group Stage Contributions on Strength for Random Treatment, Last 10 Rounds

	(1)
Strength	18.52**
	[4.940,32.10]
Constant	-3.042
	[-30.70,24.62]
Observations	480
Clusters	6

Notes: Regression of group stage contributions on strength. Random treatment only. Last 10 rounds. Wild cluster bootstrapping 95% confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 12: Mean Expected Payoff, Last 10 Rounds

	Mean	Equilibrium
Increasing	325.036	332.91
	[293.782, 349.353]	
Decreasing	344.885	370.81
	[327.807, 367.983]	
Symmetric	320.917	356.25
	[312.527, 332.746]	
Random	330.990	350.83
	[311.452, 348.806]	
Observations	1920	
Clusters	24	

Notes: Mean expected payoff by treatment with wild cluster bootstrapping 95% confidence intervals in brackets. Last 10 rounds.

Table 13: Mean Expected Payoff, All Rounds

	Mean	Equilibrium
Increasing	315.440	332.91
	[282.801, 346.466]	
Decreasing	345.195	370.81
	[326.628, 365.937]	
Symmetric	320.252	356.25
	[307.875, 334.834]	
Random	325.884	350.83
	[296.873, 352.881]	
Observations	3840	
Clusters	24	

Notes: Mean expected payoff by treatment with wild cluster bootstrapping 95% confidence intervals in brackets. All rounds.

Table 14: Mean Within-Group Total Contributions, Last 10 Rounds

	Mean	Equilibrium
Increasing	362.754	384.18
	[304.125, 446.582]	
Decreasing	356.413	308.38
	[281.961, 426.815]	
Symmetric	413.850	337.50
	[376.688, 446.219]	
Random	355.800	341.60
	[303.686, 415.649]	
Observations	960	
Clusters	24	

Notes: Mean within-group total contributions by treatment with wild cluster bootstrapping 95% confidence intervals in brackets. Last 10 rounds.

Table 15: Mean Within-Group Total Contributions, All Rounds

	Mean	Equilibrium
Increasing	386.921	384.18
	[313.435, 464.836]	
Decreasing	345.098	308.38
	[283.103, 405.438]	
Symmetric	410.840	337.50
	[362.454, 444.097]	
Random	365.765	341.60
	[289.291, 446.855]	
Observations	1920	
Clusters	24	

Notes: Mean within-group total contributions by treatment with wild cluster bootstrapping 95% confidence intervals in brackets. All rounds.

Table 16: Group Stage Contributions on Lagged Variables

	Symmetric	Increasing	Decreasing	Random
Own Group Contribution	0.589***	0.671***	0.777***	0.462***
	[0.359, 0.887]	[0.363, 0.811]	[0.529, 0.885]	[0.344, 0.536]
Partner Group Contribution	0.049	0.023	0.023	0.048
	[-0.012, 0.103]	[-0.019, 0.099]	[-0.027, 0.145]	[-0.061, 0.107]
Opponent Group Contribution	0.038	0.053**	0.042***	0.099***
	[-0.012, 0.095]	[0.002, 0.105]	[0.028, 0.058]	[0.054, 0.126]
Won Group Stage	-2.438	3.781*	-2.950	-0.309
	[-5.909, 1.052]	[-0.722, 8.301]	[-9.488, 2.967]	[-9.700, 10.116]
Won Individual Stage	6.117**	2.078	4.738**	2.337
	[2.277, 10.027]	[-2.689, 7.004]	[1.250, 10.524]	[-3.904, 9.213]
Observations		3648		
Clusters		24		

Regression of group stage contributions on lagged variables. Round 1 data omitted. Wild cluster bootstrapping 95% confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 17: Individual Stage Contributions on Lagged Variables

	Symmetric	Increasing	Decreasing	Random
Own Individual Contribution	0.438**	0.480***	0.513***	0.368***
	[0.300, 0.675]	[0.384, 0.639]	[0.298, 0.788]	[0.221, 0.518]
Partner Individual Contribution	0.082**	0.075**	-0.002	0.052
	[0.002, 0.132]	[0.010, 0.125]	[-0.088, 0.113]	[-0.057, 0.138]
Won Group Stage	-20.388	-6.797	-14.284*	-15.759**
	[-45.815, 7.564]	[-21.465, 7.762]	[-29.989, 1.798]	[-30.522, -4.311]
Won Individual Stage	15.701	10.716	-9.957	-3.047
	[-11.179, 38.556]	[-13.452, 37.562]	[-37.182, 24.708]	[-24.748, 12.666]
Current Group Contribution	-0.171	0.151	-0.036	0.097
	[-0.407, 0.182]	[-0.036, 0.549]	[-0.371, 0.418]	[-0.157, 0.427]
Observations	3648			
Clusters	24			

Regression of individual stage contributions on lagged variables controlling for current period group stage contributions. Round 1 data omitted. Wild cluster bootstrapping 95% confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

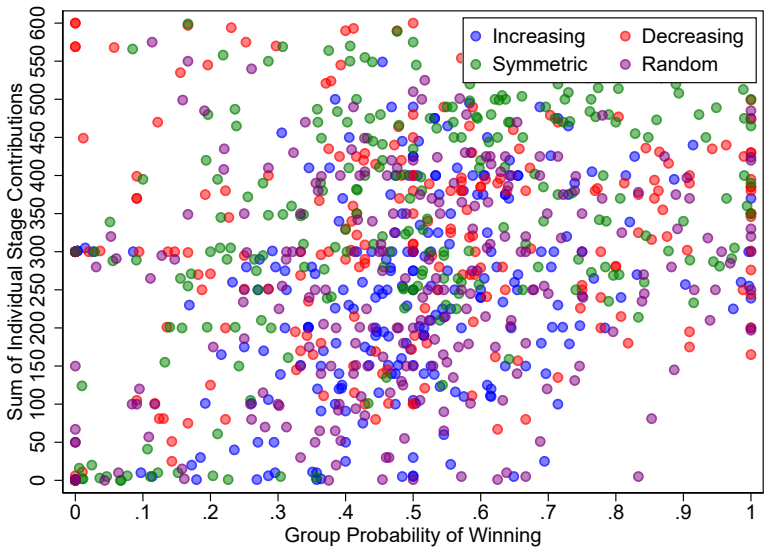


Figure 15: Sum of Ind. Stage Contributions over Group Probability of Winning, Last 10 Rounds

Table 18: Individual Stage Contributions and Controls, All Rounds

	(1)	(2)	(3)	(4)	(5)	(6)
Exogenous	39.569**	40.047**	39.569**			
	[13.678,65.459]	[13.716,66.377]	[13.675,65.463]			
Increasing	1.827	1.581	1.827	1.984	-2.500	0.607
	[-21.330,24.984]	[-20.916,24.079]	[-21.333,24.987]	[-21.870,25.839]	[-25.707,20.707]	[-24.032,25.246]
Decreasing	23.952	24.914	23.952	24.494	19.203	20.465
	[-3.834,51.738]	[-3.354,53.183]	[-3.837,51.741]	[-4.475,53.463]	[-9.758,48.163]	[-9.019,49.950]
Group contribution		0.028				
		[-0.138,0.194]				
Group probability of winning			77.541***			
			[55.299,99.784]			
Relative Strength				8.604		
				[-3.970,21.178]		
Strength Ratio					66.063**	297.029*
					[26.816,105.309]	[56.922,537.136]
Strength Ratio Squared						-175.898
						[-360.299,8.502]
Constant	123.225***	121.551***	84.454***	112.888***	80.384***	12.168
	[105.926,140.524]	[98.792,144.309]	[62.373,106.536]	[89.380,136.396]	[53.153,107.615]	[-62.142,86.477]
Observations	3840	3840	3840	2880	2880	2880
Clusters	24	24	24	18	18	18

Notes: Individual stage contribution on treatment and controls. All rounds. Wild cluster bootstrapping 95% confidence intervals in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 19: Individual Stage Contributions, All Rounds

	(1)
Increasing	-64.91***
	[-99.08,-30.73]
Decreasing	-20.10
	[-60.27,20.06]
Random	-55.65**
	[-95.39,-15.91]
Group Contribution	-0.250*
	[-0.488,-0.0115]
Increasing × Group Contribution	0.491*
	[0.0745,0.908]
Decreasing × Group Contribution	0.00703
	[-0.481,0.495]
Random × Group Contribution	0.341*
	[0.000289,0.682]
Constant	173.4***
	[148.1,198.8]
Observations	3840
Clusters	24

Notes: Individual stage contribution on treatment and group stage contributions. All rounds. Wild cluster bootstrapping 95% confidence intervals in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

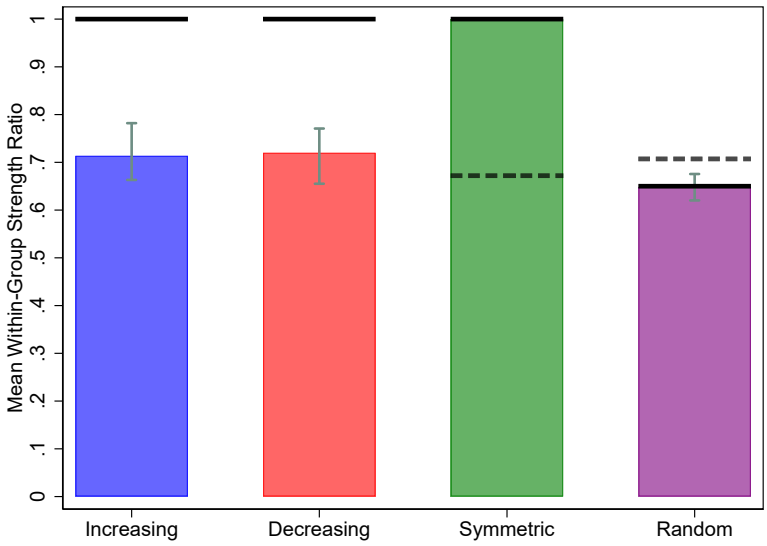


Figure 16: Strength Ratio, All Rounds

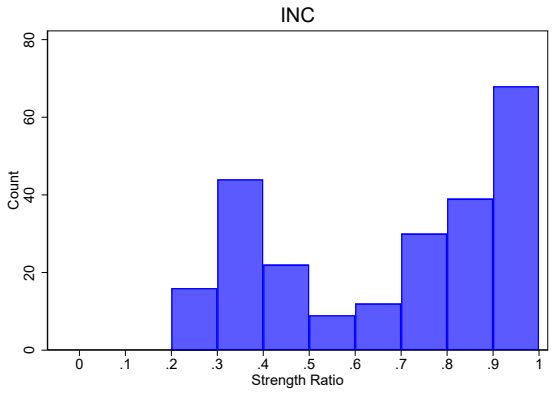


Figure 17: Histogram of Strength Ratios, Increasing, Last 10 Rounds

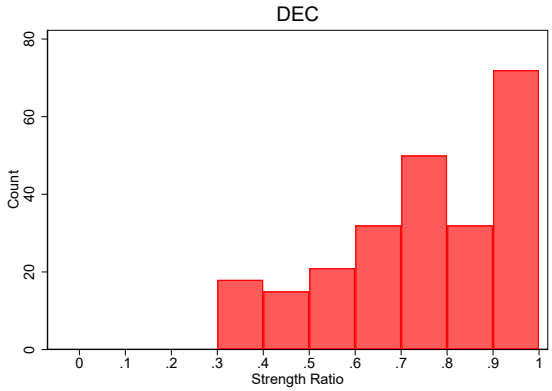


Figure 18: Histogram of Strength Ratios, Decreasing, Last 10 Rounds

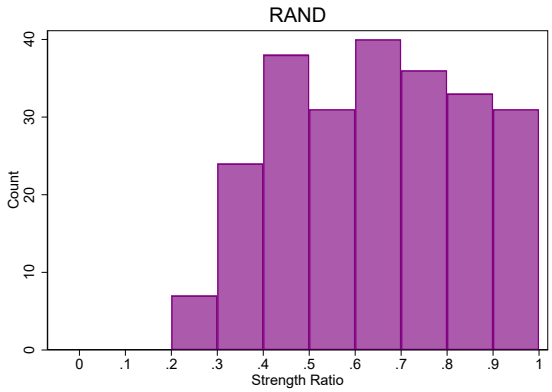


Figure 19: Histogram of Strength Ratios, Random, Last 10 Rounds

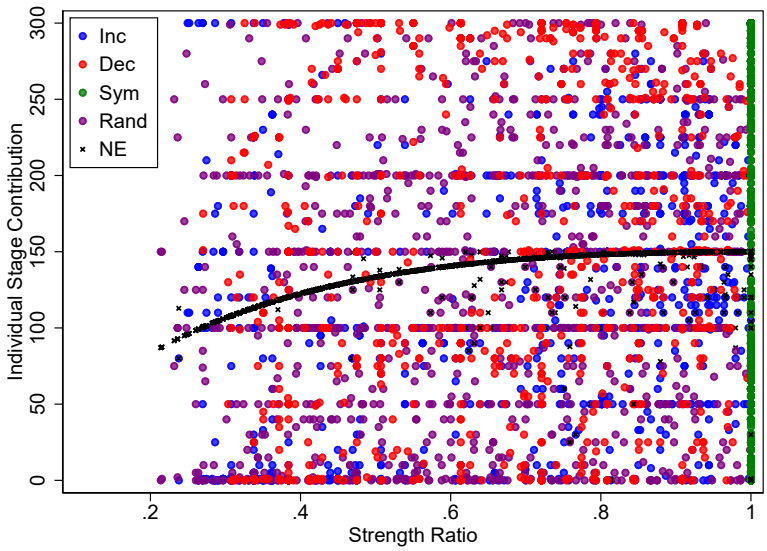


Figure 20: Individual Stage NE by Strength Ratio, All Rounds

Table 20: Individual Stage Contributions by Treatment and Type, Exact, Last 10 Rounds

Type	Increasing	Decreasing	Symmetric	Random
Equal	133.48	122.90	174.10	.
	[16.08,169.11]	[49.92,177.17]	[151.60,188.34]	
Strong	134.64	182.32	.	138.75
	[109.32,169.79]	[119.49,235.84]		[114.09,165.05]
Weak	105.03	155.18	.	112.97
	[78.40,138.59]	[112.47,205.79]		[88.95,142.33]

Notes: Estimates from regressing individual stage contribution on treatment dummy and type dummy with exact equal categorization. Last 10 rounds. 95% wild cluster bootstrap confidence intervals in brackets.

Table 21: Individual Stage Percent Deviation by Treatment and Type, Exact, Last 10 Rounds

Type	Increasing	Decreasing	Symmetric	Random
Equal	-0.11	-0.18	0.16	.
	[-0.89,0.10]	[-0.67,0.18]	[0.01,0.26]	
Strong	0.01	0.29	.	0.01
	[-0.19,0.26]	[-0.17,0.71]		[-0.16,0.21]
Weak	-0.25	0.09	.	-0.18
	[-0.46,0.06]	[-0.21,0.42]		[-0.37,0.02]

Notes: Estimates from regressing percent deviation from individual stage NE on treatment dummy and type dummy with exact equal categorization. Last 10 rounds. 95% wild cluster bootstrap confidence intervals in brackets.

Table 22: Individual Stage Contributions by Relative Strength, All Rounds

	(1)
Increasing	1.450
	[-26.59,29.49]
Decreasing	17.66
	[-31.75,67.07]
Relative Strength	6.991
	[-0.786,14.77]
Increasing × Relative Strength	0.427
	[-10.36,11.22]
Decreasing × Relative Strength	5.915
	[-43.07,54.90]
Constant	114.8***
	[94.23,135.4]
Observations	2880
Clusters	18

Notes: Regression of individual stage contributions on relative strength and treatment dummies. All rounds. Wild cluster bootstrapping 95% confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 23: Individual Stage Contributions on Strength Ratio, All Rounds

	(1)
Increasing	-35.93
	[-83.60,11.74]
Decreasing	53.19
	[-15.19,121.6]
Strength Ratio	62.34*
	[9.291,115.4]
Increasing × Strength Ratio	47.16
	[-30.78,125.1]
Decreasing × Strength Ratio	-46.81
	[-139.2,45.53]
Constant	82.80***
	[55.19,110.4]
Observations	2880
Clusters	18

Notes: Regression of individual stage contributions on strength ratio and treatment dummies. All rounds. Wild cluster bootstrapping 95% confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 24: Individual Stage Contributions on Strength Ratio, Type, and Treatment, Last 10 Rounds

	(1)
Increasing	-93.85
	[-220.6,32.89]
Decreasing	30.45
	[-63.72,124.6]
Strong	35.17
	[-17.22,87.55]
Strength Ratio	66.04
	[-34.90,167.0]
Increasing × Strong	63.42
	[-26.96,153.8]
Decreasing × Strong	-3.616
	[-276.9,269.7]
Increasing × Strength Ratio	132.6
	[-26.24,291.4]
Decreasing × Strength Ratio	12.56
	[-113.0,138.1]
Strong × Strength Ratio	-14.53
	[-76.71,47.65]
Increasing × Strong × Strength Ratio	-92.01*
	[-175.5,-8.495]
Decreasing × Strong × Strength Ratio	8.168
	[-324.7,341.1]
Constant	70.29
	[-1.094,141.7]
Observations	1314
Clusters	18

Notes: Regression of individual stage contributions on strength ratio, type and treatment dummies. Last 10 rounds. Wild cluster bootstrapping 95% confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 25: Slope Coefficients on Strength Ratio by Type, Last 10 Rounds

	Random	Increasing	Decreasing
Weak	66.043	198.608***	78.604*
	[-67.126, 185.091]	[42.470, 313.848]	[-0.715, 151.050]
Strong	51.514	92.069**	72.244
	[-47.097, 146.413]	[2.399, 161.794]	[-230.165, 355.439]
Observations		1314	
Clusters		18	

Notes: Slope coefficients on strength ratio from regression of individual stage contributions on strength ratio, type, and treatment. Last 10 rounds. Wild cluster bootstrapping 95% confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 26: Sum of Individual Stage Contributions by Strength Ratio, Last 10 Rounds

	(1)
Increasing	-108.5
	[-275.8,58.70]
Decreasing	148.2
	[-73.40,369.8]
Strength Ratio	117.6
	[-46.89,282.0]
Increasing × Strength Ratio	140.2
	[-88.31,368.7]
Decreasing × Strength Ratio	-120.9
	[-395.6,153.8]
Constant	175.7**
	[65.46,286.0]
Observations	720
Clusters	18

Notes: Regression of the sum of individual stage contributions on strength ratio. Last 10 rounds. Wild cluster bootstrapping 95% confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

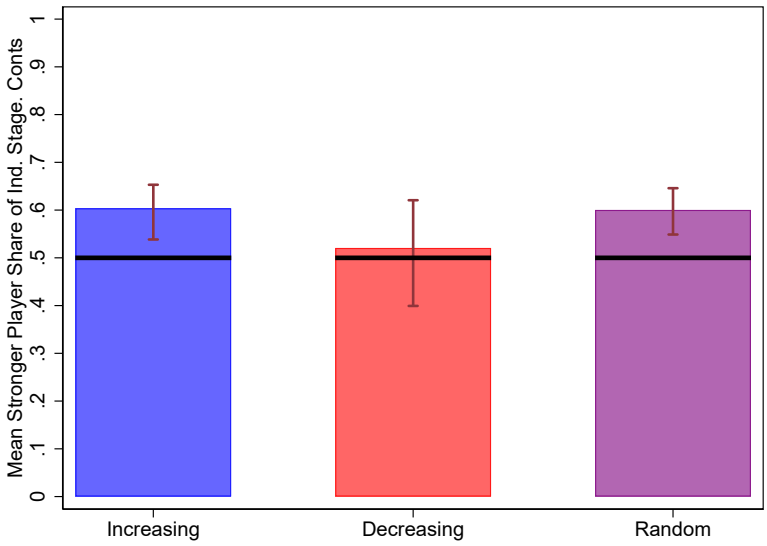


Figure 21: Share of Individual Stage Contributions, All Rounds

Table 27: Strong Player Share of Individual Stage Contributions by Strength Ratio, All Rounds

	(1)
Increasing	0.273*
	[0.0560,0.490]
Decreasing	-0.160
	[-0.601,0.281]
Strength Ratio	-0.159*
	[-0.301,-0.0170]
Increasing × Strength Ratio	-0.386**
	[-0.630,-0.141]
Decreasing × Strength Ratio	0.125
	[-0.387,0.638]
Constant	0.704***
	[0.582,0.825]
Observations	1331
Clusters	18

Notes: Regression of stronger player share of individual stage contributions on strength ratio and treatment dummies. All rounds. Wild cluster bootstrapping 95% confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

C Experiment Instructions (Increasing Treatment)

Introduction

Thank you for participating in today's experiment. You will be participating in a type of market in which your earnings will depend on your own decisions and the decisions of other participants. Your total earnings, including your $7 show-up fee, will be paid to you privately at the end of the experiment. Please remain quiet and do not communicate with other participants.

We would not want anyone's earnings to suffer because you did not understand the rules of how these markets operate. If you have a question or a problem during the experiment, we encourage you to raise your hand and one of the experimenters will do their best to try and answer it for you. We cannot tell you what a good or bad decision is for you, but we will gladly help explain the rules.

In the experiment, your payoffs will be denominated in points. We will explain in a moment how points are converted into U.S. dollars. All amounts in this experiment are expressed in points. The exchange rate is 1 point = $0.01. This part of the experiment consists of a sequence of 20 decision rounds.

Pairing and matching

At the beginning of the experiment, you will be randomly placed into a pool of 8 participants. You will remain in this pool for the duration of the experiment, and you will only interact with the other members of your pool.

Within each round, you will be paired in a group with one other participant. Your group will then be matched with another group of two different participants. Between rounds, you'll be randomly re-paired with another group member and randomly rematched with another group.

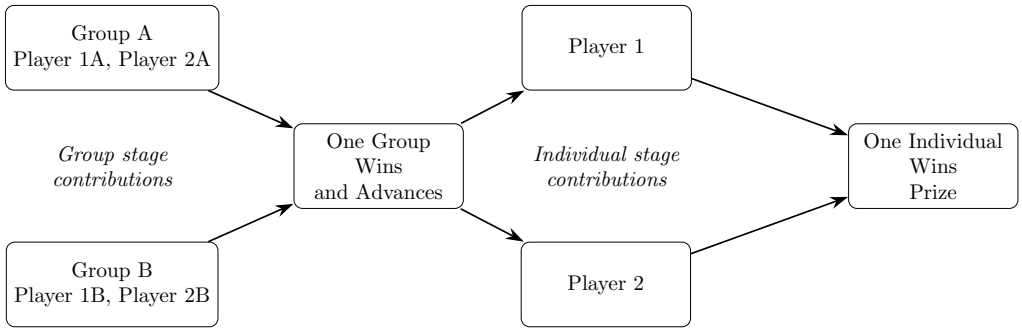
Endowment

At the beginning of each round, you will be given an endowment of 300 points. You may use any number of these points to make decisions in a given round.

Round Overview

Each round consists of two stages: the group stage and the individual stage. In the group stage, two groups will compete to determine which group advances to the individual stage. In the individual stage, members of the winning group will compete with each other for a prize. Your task will be to contribute points from your endowment to the competition in each stage. Ultimately, only one out of the four players that enter a match in each round will win the prize. This player must belong to the winning group and win the individual stage.

The following flow chart summarizes how each round will proceed:



Payoffs

The payoff from each round will depend upon the outcomes in both stages in the following way:

$$\text{Round payoff} = \underbrace{300}_{\text{Endowment}} - \text{group stage contribution} - \underbrace{\text{individual stage contribution}}_{\text{only if group wins}} + \underbrace{600}_{\substack{\text{Prize} \\ \text{only if group wins} \\ \& \text{ individual wins}}}$$

In each round, you earn your endowment of 300 points minus the points you contribute to the group stage. If your group wins, you also pay the points you contribute to the individual stage. Finally, if you are the one individual who wins the prize, you receive 600 points.

Group Stage

In the group stage, two groups of two players will compete to determine which group advances to the individual stage. You will decide how many points you wish to contribute to your group. You can contribute any integer number between 0 and 300 (0 and 300 inclusive). Any points which you don't contribute in this stage will remain with you for the next stage. Your group's total contributions will be the sum of the contributions of you and your group mate.

You can't earn any additional points in the group stage. You compete only for the right to advance to the individual stage where you will have the opportunity to compete for a prize.

What is the likelihood that your group wins?

Only one of the two groups can win. Your group's likelihood of winning is computed as the share of your group's contributions in the total contributions from both groups. The **probability that your group wins** is given by:

$$\frac{\text{your group's contributions}}{\text{your group's contributions+other group's contributions}}$$

If all players contribute 0 points, then the probability that your group wins is $\frac{1}{2} = 50.00\%$.

Once every player has submitted their contributions, the computer program will use the probabilities of winning to determine which group advances to the individual stage.

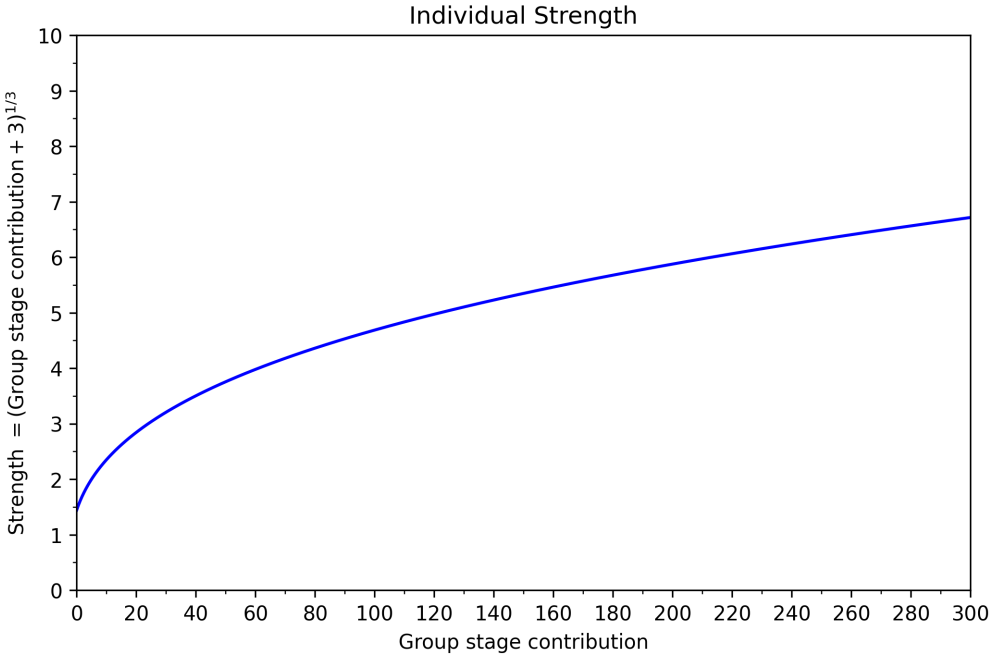
Individual Stage

In the individual stage, you will compete with your group mate for a prize of 600 points. At this point, you won't know whether your group won or lost in the group stage. You will make your decision in the individual stage as if your group won.

You will choose how many points you wish to contribute to the individual competition. You can choose any integer number less than or equal to your **remaining endowment**. Your **remaining endowment** is the share of your endowment that you didn't contribute in the group stage (300 − group stage contribution). The more points you contribute in the group stage, the fewer points you have to contribute in the individual stage.

Individual strength and multiplied contribution

Your contribution in this stage will be multiplied by your individual **strength** which depends on your group stage contribution. The following figure illustrates how group stage contribution determines strength:



Notice, the *more* points you contribute in the group stage, the *higher* your strength will be in the individual stage. Your group mate's individual stage contributions will be multiplied by his/her strength which depend on his/her group stage contributions in the same way.

Your multiplied contribution is the product of your individual stage contribution and your strength:

$$\text{Multiplied contribution} = \text{your individual stage contribution} \times \text{your strength}$$

What is the likelihood that you win the individual stage? Only one of you or your group mate can win the individual stage. The likelihood that you win is the share of your multiplied contribution in the total multiplied contributions of both you and your group mate. The **probability that you win** is given by:

$$\frac{\text{Your multiplied contribution}}{\text{Your multiplied contribution} + \text{your group mate's multiplied contribution}}$$

If you and your group mate both invest zero, then the probability that you win is:

$$\frac{\text{Your strength}}{\text{Your strength + your group mate's strength}}$$

Once every player has submitted their contributions, the computer program will use the probabilities of winning to determine which individual wins the prize.

How are your earnings determined?

You will participate in a series of 20 decision rounds. Before the experiment began, **four** of these rounds were chosen randomly (with all rounds being equally likely to be chosen) to determine your earnings. At the end of the experiment, you will be informed about which four rounds were chosen and your payoff from each of those four rounds. Then your earnings from the experiment will be the sum of your payoffs from the four randomly selected rounds.